MIL-F-48704B(AR)
6 JULY 1982
SUPERSEDING
MIL-F-48704A(AR)
30 NOVEMBER 1981

MILITARY SPECIFICATION

FUZE SETTER: M38

This specification is approved for use by USA Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 General. This specification covers the requirements and test limits for the M38 Fuze Setter. The M38 Setter is designed for use with the M735 Fuze which is a part of the M753 8-inch Nuclear Projectile. The specification covers the complete M38 setter assembly including outer case and power cable.

1.2 Safety Requirement. This specification is intended to assure a level of performance for the fuze setter which is consistent with the overall weapon system reliability and safety requirements. Because of the stringent inflight nuclear safety requirements on the weapon system, manufacturing acceptance testing of the setter must be adequate to support an extremely low probability of producing an unsafe fuze setting which is not apparent to the setter operator. This implies that all design features and self tests which bear on this capability be thoroughly tested. Furthermore, acceptance tests must assure each setter's ability to sustain certain critical internal faults (e.g., fault detection circuit and time base errors) without violating the display validity in the safety sense. This is accomplished by artificially inducing such faults within the setter to demonstrate that each setter can in fact tolerate these potential safety faults by either becoming inoperative, or by warning the operator with a FUZE STATUS display which is invalid for the intended mission.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Armament Research and Development Command, ATTN: DRDAR-TST-T, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Method of
MIL-A-2550	Ammunition and Special Weapons, General Specifications for
MIL-F-48703	Fuze, Proximity: M735
MIL-Q-50829	General Quality Assurance Provisions for Proximity Fuzes and Related Components
MIL-P-55110	Printed Wiring Boards

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-1275	Characteristics of 28-volt DC Electrical Systems in Military Vehicles

DRAWINGS

USA ERADCOM, Harry Diamond Laboratories (FSCM 19202)

DL11727881	Data List for FUZE Setter: M38, PN 11727881
EDL11727881	Inspection Equipment List for Setter
11710170	Interface, Fuze Setter Schematic

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Samples. The contractor is responsible for all requirements for providing and testing of samples as prescribed herein. When this specification is assigned to a subcontractor, it shall be the prime contractor's responsibility to insure that sample requirements are met. The prime contractor shall notify the Government at the time of assignment to a subcontractor whether the testing will be performed by the prime or by the subcontractor, as well as the method of control by the prime contractor.

3.1.1 First article approval sample. Prior to the start of regular production, except when production on a new contract at the same facility follows within 90 days production of acceptable material under this specification, the contractor shall manufacture and submit a first article approval sample (FAAS) of two units using as far as possible the methods and processes proposed for quantity production. When processes differ from the production processes, they shall be basically equivalent and the difference

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

shall be fully described in documentation submitted with the first article sample. The sample shall be tested in accordance with paragraph 4.4.5.2. Additional samples that may be required because of failure of the sample to meet the requirements of this specification shall be supplied by the contractor at his expense. Prior to approval of the first article approval sample, acquisition of parts and materials or initiation of production will be at the sole risk of the contractor.

3.1.2 Supplemental samples. At the Government's discretion, a supplemental sample may be required whenever there is a break in production continuity of 90 days or more, or a change in design, material or process that may affect safety, operability, reliability or interchangeability as defined by the Requirements (3.) and Classification of Defects (4.4.4). In all cases, such changes will require prior approval of the procuring activity.

3.1.3 Comparison sample. At least one or more accepted setters as specified by the procuring agency may be tested to demonstrate that the first article approval sample requirements continue to be met. (See 6.1.3)

3.2 Construction. The device shall be constructed in accordance with the applicable drawings (DL11727881) to the extent prescribed herein.

3.2.1 Materials. Materials shall be those specified by the applicable drawings.

3.2.2 Dimensions and technical notes.

3.2.2.1 Listed. Those dimensions and technical requirements listed in the Classification of Defects are mandatory.

3.2.2.2 Unlisted. The contractor may propose changes to characteristics shown on the drawings, but not listed in the Classification of Defects, for the purpose of adapting the item to established manufacturing practices. Such proposals must be accompanied by evidence that the change does not affect the function of the item and that all requirements will be met. If the Government confirms the contention of the contractor, the change will be approved for the duration of the contract. In case of dispute, the characteristics of the drawings shall apply. Approval of a change under provisions of this paragraph does not relieve the contractor from establishing and maintaining an adequate quality assurance program as elsewhere required. Provisions of this paragraph shall not be used to obtain approval for use of discrepant material (i.e., produced before approval is obtained); or for design changes, which should be requested in accordance with change provisions of the contract document.

NOTE

It is expected that the above changes will normally be accomplished prior to the construction of the First Article Approval Sample and reflected therein.

3.2.2.3 Components. Components with lower failure rates, or closer tolerances on the component values, or higher power dissipation ratings may be used in place of those called out on the drawings.

Source: https://assist.dla.mil Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3 Performance requirements. The requirements contained herein are mandatory. Any requirements stated in this section, but not explicitly tested for in paragraph 4.6, are still required to be met. Unless otherwise specified, the following requirements apply to the assembled M38 Fuze Setter over the range of voltage, temperature, and environmental conditions specified herein. All operating requirements apply to the setter when powered by either (1) external vehicular power, or (2) external G76 generator power.

3.3.1 Functional requirements. The following specifications apply to the chassis assembly (11735215) providing access to test connector J7 and powered from sources simulating external vehicular power and G76 generator power. All references to the fuze herein refer to the M735 Fuze or the M735 Fuze Nose Section as specified by MIL-F-48703. Tests relating to these specifications shall be performed with either a fuze or simulated fuze circuits as specified in 4.5.5 and 4.6.1. The setter shall be subjected to the range of simulated fuze outputs, including faults, specified herein.

3.3.1.1 Power supply. The requirements specified herein apply to the setter when powered from an external power source (4.5.4) connected to the setter's external power connector.

3.3.1.1.1 Input voltage. The setter shall operate properly as specified in 3.3.1 when powered externally with an input voltage of 17V dc minimum and 33V dc maximum.

3.3.1.1.2 Input current. The setter dc current drain from the external power source shall not exceed 2.5 amperes during all setter operating sequences.

3.3.1.1.3 Input voltage regulation. The input regulator output voltage at connector J6 shall be 10.9 ± 1.0 volts dc when the external input voltage is in the range specified in 3.3.1.1.1 with the setter power on and not sequencing.

3.3.1.1.4 Low voltage indicator. The low voltage indicator "B" shall be displayed when the voltage at J6 is decreased to 9.0 ± 0.2 Vdc and remain on thereafter even if the voltage at J6 is increased. In addition, when the input voltage reaches the "B" threshold, the SELECT display shall be reset to "0." and the STATUS display sequence halted.

3.3.1.1.5 Power shutdown threshold. Power to the setter shall be internally disabled when the voltage at J6 drops to 8.0 ± 0.2 Vdc.

3.3.1.1.6 5 volt regulation. The voltage at connector J7-12 shall be 5.0 ± 1.0 Vdc with setter power on and sequencing.

3.3.1.2 Vehicle power noise immunity. The setter shall operate properly as specified in 3.3.1 (except as noted in 3.3.1.2.3 and 3.3.1.2.4) when external power line noise (simulating a military vehicle power source as specified by MIL-STD-1275) is injected at J1-S along with the steady state dc voltage. The noise simulation signals shall consist of the following external power supply perturbations superimposed on the specified dc voltage and separately applied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.2.1 Ripple. A periodic waveform that is swept from 50 Hz to 200 KHz in a 2 second (± 10%) interval will be applied to a setter load at a dc operating voltage of 25.0V. The amplitude of this waveform is 14V P-P (± 10%) when applied to a 16.5 ohm load (± 5%).

3.3.1.2.2 Spikes. A pulse train consisting of pulses with each pulse containing a minimum of 15 millijoules of energy which when combined with the dc operating voltage have a minimum peak of ± 250V. The pulse interval shall be 1 second (± 10%) between pulses.

3.3.1.2.3 Surges. The test simulates a vehicular power supply operating with either a no-fault condition or with a single fault condition as defined by MIL-STD-1275.

3.3.1.2.3.1 No fault condition. The external power source shall be decreased in a step transition, from an operating level of 28 VDC to 18 VDC maximum for a duration of 2 milliseconds (± 10%) and again for a duration of 1 second (± 10%). The external power source shall be increased in a step transition, from an operating level of 28 VDC to 40 VDC maximum for a duration of 2 milliseconds (± 10%), and again for a duration of 50 milliseconds.

3.3.1.2.3.2 Single fault condition. The external power source shall be decreased in a step transition from an operating level of 28 VDC to 15 VDC maximum for a duration of 2 milliseconds (± 10%) and again for a duration of 1 second (± 10%). The external power source shall be increased in a step transition, from an operating level of 28 VDC to 100 VDC minimum for 50 milliseconds (± 10%). The setter is not required to be operational through the 15 VDC/100 VDC surges, however it shall survive and be operational following the surge and it shall not produce a fault through the 15 VDC/100 VDC surge which would allow an operator to accept an incorrectly set fuze.

3.3.1.2.4 Vehicle starting disturbance. A simulated starting power perturbation shall be applied which drops the dc operating level to 6V (± 10%) in a step transition with a simulated exponential recovery which begins immediately and requires 1 second (± 10%) to reach a level of 16V dc (± 10%) where the voltage remains for 30 seconds minimum, then recovers to the dc quiescent level. The setter is not required to be operational through this simulated disturbance, however it shall not produce a fault which would allow an operator to accept an incorrectly set fuze.

3.3.1.3 G76 noise immunity. The setter shall operate as specified in 3.3.1 (except 3.3.1.2) when powered by the output of a G76 generator at a cranking speed which corresponds to a level above the power shutdown threshold (3.3.1.1.5), but below the "B" indicator threshold (3.3.1.1.4) while the setter is sequencing. This requirement simulates the fault condition of a "B" circuit failure, therefore, the "B" circuit shall be disabled to demonstrate G76 setter low voltage compatibility specified herein.

3.3.1.4 Setter self tests. The setter shall have the self test capabilities specified below. In addition, the self test outputs shall be detected on test connector J7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.4.1 169 prejam generator. The setter self test fault indicator (F) shall be displayed and the set sequence (3.3.1.8.2) inhibited when the 169 count prejam generator circuit in the setter is externally disabled at the setter test connector J7.

3.3.1.4.2 Decade test counter. The setter F indicator shall be displayed and the set sequence inhibited when the decade test counter in the setter is externally disabled at the test connector J7.

3.3.1.4.3 Erase voltage output.

3.3.1.4.3.1 Amplitude. The setter F indicator shall be displayed and the set sequence inhibited if any of the three erase pulse output amplitudes at J3-S are less than 25.0V dc. The F indicator shall not be displayed when all three erase pulse amplitudes exceed 27.0V dc.

3.3.1.4.3.2 Error detection circuit. The setter F indicator shall be displayed and the set sequence inhibited when the erase voltage amplitude error detection circuit is externally disabled at the setter test connector J7.

3.3.1.4.4 Fuze/setter continuity.

3.3.1.4.4.1 Continuity fault detection. The setter shall accept a fuze setter continuity loop resistance of 2 ohms or less and always detect a continuity fault if the loop resistance exceeds 5 ohms. When a continuity fault is detected, the continuity fault indicator C shall be displayed and the fault detected on test connector J7.

3.3.1.4.4.2 Power disable. Power from the setter to the fuze being set shall not be enabled when a continuity fault as specified in 3.3.1.4.4.1 is detected.

3.3.1.5 Keyboard entry/display. The setter shall accept any time entry from 13.0 to 200.0 seconds inclusive in 0.1 second increments followed by a proper one digit (1, 2, or 3) option entry. The entry shall be accomplished by successive operator keystrokes which shall be immediately displayed and retained on the SELECT display until setter power is removed, or until the CE key is pressed.

3.3.1.6 Entry error detection. The setter shall detect the invalid data entries specified below and inhibit setter operation until the CE key is pressed and a valid entry is completed. In addition, these errors shall be detected on test connector J7.

3.3.1.6.1 Time error. The setter shall display the entry error indicator E and inhibit further keyboard entry when a time selection of less than 13.0 seconds or greater than 200.0 seconds is entered.

3.3.1.6.2 Option error. The setter shall display the E indicator and inhibit the set sequence (3.3.1.8.2) when an option selection of 0 or greater than 3 is entered.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.7 Error display priority. In the event more than one of the possible setter/operator errors coexist, the setter shall display the highest priority error according to the following:

Priority	Error	Display Code
High	Self Test Fault	F
	Low Voltage	B
	Continuity	C
Low	Entry Error	E

3.3.1.8 Fuze memory setting, verification, and interrogation.

3.3.1.8.1 Fuze counter prejam. The setter shall provide prejam counts to the fuze timing memory counters prior to injecting the counts corresponding to the set time. The fuze option memory counters shall be prejammed to the counter state corresponding to the desired option memory output.

3.3.1.8.1.1 Time prejam. The setter prejam outputs to the fuze timing counters (via the Setter Clock input lines on J3-S) shall be 169 counts which shall include 1 count for fuze battery activiation delay and 1 additional count for fuze memory read delay.

3.3.1.8.1.2 Option prejam. The setter prejam outputs to the fuze option counters (via the MNOS lines on J2-S) shall be as specified below:

Option 1 - 226 counts
Option 2 - 258 counts
Option 3 - 292 counts

3.3.1.8.2 Test/set/verify sequence. The fuze test (3.3.1.10), set, and playback verification sequence shall be automatically initiated at the completion of a valid data entry. The entire sequence shall be automatic and is complete when an accepted fuze (as specified in MIL-F-48703) or a simulated fuze playback of the entry is displayed on the FUZE STATUS LED panel. A properly operating and safe fuze shall produce a time-option status display corresponding to the SELECT display (see 3.3.1.8.2.3); a defective or unsafe fuze shall produce a FAIL in the FUZE STATUS display (see 3.3.1.10).

3.3.1.8.2.1 Time setting accuracy. The ability of the setter to accurately set a fuze shall be such that the set fuze real time playback time agrees with the selected time within ± .075 second for all set times from 13.0 to 125.0 seconds. The fuze playback signal shall be the 2-out-of-3 decoded time output from the 120-count decode counters in each fuze channel; the external counter time shall be increased by 0.100 second to compensate for the one count prejam bias for fuze battery activation delay.

3.3.1.8.2.2 Option setting accuracy. The option memory outputs from a fuze which has been set by the setter and monitored shall be in exact agreement with the selected option (no extra or missing outputs) for each of the three option choices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.8.2.3 Time verification accuracy. The playback verification accuracy after setting shall be such that when the setting error (3.3.1.8.2.1) is combined with the verification error in detecting fuze memory playback time, the resulting FUZE STATUS display time shall agree with the keyboard data entry and SELECT display to within ± 0.1 second. The fuze playback signal shall be the 2-out-of-3 decoded output from the three simulated fuze decode counters (120 count) in the setter.

3.3.1.8.2.4 Option verification accuracy. The fuze option memory output displayed during the playback verification sequence shall be in exact agreement with the memory outputs detected from that fuze by either independent external measurement of these output functions, or monitoring of the interface of a simulated fuze, for each of the three option choices.

3.3.1.8.2.5 Test/set/verify cycle time. The time required to complete the automatic test, setting, and verification sequences following keyboard data entry shall not exceed 12.0 seconds for any set time from 13.0 to 200.0 seconds.

3.3.1.8.3 Interrogate sequence. The fuze memory interrogate sequence shall be initiated by the unique two-keystroke entry .0 (decimal-zero) after the CE key is pressed. The complete interrogate cycle shall include the fuze tests (3.3.1.10) and fuze memory playback in an automatic single sequence. Fuze setting, and setter comparison tests between fuze outputs and select entry data, are omitted in the interrogate sequence. The interrogate sequence is complete when the fuze memory contents are displayed on the FUZE STATUS LED panel. A properly operating fuze shall produce a time-option FUZE STATUS display which corresponds to the fuze memory contents; a defective fuze shall produce a FAIL X in the FUZE STATUS display where X is a single digit number code corresponding to the specific fuze fault detected (see 3.3.1.10).

3.3.1.8.3.1 Interrogate time accuracy. The accuracy of the setter in the interrogate mode to display the time set into the fuze shall be such that the displayed time agrees with the known set time within 0.100 second. The known set time shall be based on independent external measurement of fuze 2-out-of-3-time-out from the 120 count decode counters in each fuze channel, using a fuze power source with a rise time of less than 1 millisecond. The external counter time shall be increased by 0.1 second to compensate for the 1 count prejam bias for fuze battery activation delay. In addition, when the interrogate sequence is performed immediately following the setting sequence (3.3.1.8.2), the verification accuracy requirement specified in 3.3.1.8.2.3 shall apply (relative to the data entry) to the interrogate sequence also.

3.3.1.8.3.2 Interrogate option accuracy. The option memory fuze outputs displayed during the interrogate sequence shall be as specified in 3.3.1.8.2.4.

3.3.1.8.3.3 Interrogate cycle time. The time required to complete the automatic interrogate sequence following keyboard entry of the interrogate command shall not exceed 10 seconds for any set time between 13.0 and 200.0 seconds.

3.3.1.8.4 Setter time base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.8.4.1 Accuracy. The error in the setter time base shall not exceed 100 ppm from a nominal 1.0240 MHz at +25°C ± 5°C.

3.3.1.8.4.2 Temperature coefficient. The temperature coefficient of the setter's internal time base shall not exceed 3.0 ppm/°C over the temperature range of -35°C to +55°C.

3.3.1.9 Fuze/setter interface. The interface specifications below apply to the setter when its outputs are loaded by the simulated fuze impedance levels specified in 4.5.3.

3.3.1.9.1 Fuze power. The setter shall provide 33.0 ± 5V dc at 60 ma max to the Fuze Power input lines on J3-S. The rise time of the power application to 20V dc shall not exceed 50 milliseconds.

3.3.1.9.2 Erase pulse. The erase pulse output to each fuze timing and option channel on J3-S shall be +28 ± 1 V dc in amplitude for 200 ± 50 ms duration applied when setter power to the fuze is 0 + 0.5V dc.

3.3.1.9.3 Memorize pulse. The memorize pulse output to each fuze timing and option channel on J3-S shall be a negative pulse generated by connecting the Erase/Memorize line of each fuze channel (4.5.5.9) to fuze common through a resistance of less than 1500 ohms for 200 ± 100 ms duration while the fuze is powered by the setter (3.3.1.9.1). The Erase/Memorize line shall not exceed 1.4V dc during the memorize pulse time.

3.3.1.9.4 Read bias voltage. The read bias voltage on the Erase/memorize line of each fuze channel shall be 16.05 ± .25V dc when fuze power is applied, except during the memorize pulse.

3.3.1.9.5 Set control signal. The Set Control signal to each fuze channel of J2-S shall be a logic voltage with a high state level of 12.8 ± 1.5 V dc and a low level not to exceed +1.0 V dc.

3.3.1.9.6 Time set clock signal. The time prejam and time set clock signals from the setter (Setter Clock) to each fuze timing channel on J3-S shall be squarewave signals with a nominal duty cycle of 50% (± 10%).

3.3.1.9.6.1 Frequency - time prejam sequence. The Setter Clock signal to each fuze channel during the time prejam shall have a frequency of 1.28 kHz ± 50 Hz.

3.3.1.9.6.2 Frequency - setting sequence. The Setter Clock signal to each fuze channel during setting shall have a frequency equal to that fuze channel's oscillator (4.5.5) input divided by 8.

3.3.1.9.6.3 Frequency - verification sequence. During the playback verification sequence, each Setter Clock frequency shall be equal to the fuze oscillator (4.5.5) input divided by 32.

3.3.1.9.6.4 Amplitude. Each Setter Clock signal shall have a high level amplitude of 12.8 ± 1.5V dc and a low level not to exceed + 1.0 V dc.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.9.7 Option set clock signal. The option clock prejam signals applied to the fuze MNOS output lines (option set clock input) on J2-S shall be rectangular wave signals with a nominal duty cycle of 50% (± 0%).

3.3.1.9.7.1 Frequency. The option clock signal frequency to each fuze option channel shall be 1.28 kHz ± 50 Hz.

3.3.1.9.7.2 Amplitude. Each option clock signal shall have a high level amplitude of 11.7 ± .9V dc and a low level not to exceed + 1.0V dc.

3.3.1.9.8 Fuze connector resistance. The input load resistance presented to fuze outputs on setter connectors J2-S and J3-S shall be as specified below measured from each connector pin to fuze/setter common, pins 6-16 of J2-S.

Connector/Pin	Resistance (±10%)	Connector/Pin	Resistance (±10%)
J2-S/2 (MNOS A)	273K	J2-S/15 (03 set B)	56K
3 (MNOS B)	273K	17 (02 set B)	56K
5 (03 set A)	56K	18 (Clock B)	56K
7 (02 set A)	56K	19 (01 set B)	56K
9 (01 set A)	56K	20 (13 V C)	100K
10 (13 V B)	2.2K	21 (Clock A)	56K
11 (13 V A)	2.2K	J3-S/1 (OSC C)	56K
12 (Clock C)	56K	8 (OSC B)	56K
13 (MNOS C)	273K	11 (OSC A)	56K

3.3.1.10 Fuze tests and fault detection. The setter shall perform the following tests on the fuze and detect simulated fuze faults within the measurement accuracy specified. If any of the following fuze faults are detected by the setter, the setter shall inhibit its automatic setting sequence and reject the fuze by displaying FAIL on the FUZE STATUS panel.

3.3.1.10.1 Fuze battery fault. The setter shall accept (no fault detected) a fuze battery resistance of 1.0M ± 0.2M ohms, and shall always detect a fault if the resistance is less than 500K or greater than 1.5M ohms. When a battery fault is detected, the FUZE STATUS panel shall display FAIL in the test/set/verify sequence (3.3.1.8.2), and FAIL 4 in the Interrogate sequence (3.3.1.8.3).

3.3.1.10.2 Fuze current fault. The setter shall accept a fuze current of 100 mA or less, and shall always detect a fault if the current is 150 mA or greater. When a fuze current fault is detected, the FUZE STATUS panel shall display FAIL in the test/set/verify sequence, and FAIL 5 in the interrogate sequence.

3.3.1.10.3 Fuze clock fault. The setter shall accept fuze clock operation if the period of each fuze clock channel is 94 ms to 106 ms and there are three clock outputs, and shall always detect a fault if any one or more clock outputs are either missing or have a period equal to or outside the range from 92-108 ms. When a clock fault is detected, the FUZE STATUS panel shall display FAIL in the test/set/verify sequence, and FAIL 6 in the interrogate sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.10.4 Fuze coincidence fault. The setter shall accept fuze timing coincidence operation if all three fuze channel timing outputs are present, and each output occurs within 8 counts of the other two channel outputs; the setter shall detect a fault if any one channel is either not present, or occurs 9 or more counts from the first of the other two outputs. When a coincidence fault is detected, the FUZE STATUS panel shall display FAIL in the test/set/verify sequence, and FAIL 7 in the interrogate sequence.

3.3.1.10.5 Fuze accuracy fault. The setter shall accept the timing accuracy of a fuze if the timing outputs from any two channels are decoded within 1 count of the selected time, and shall detect a fault if any two fuze channel outputs differ by 2 counts or greater from the selected time. When an accuracy fault is detected in the test/set/verify sequence, the FUZE STATUS panel shall display FAIL. In the Interrogate sequence, the setter FUZE STATUS display shall read the actual 2-out-of-3 decoded fuze time for either the accept or fault condition. (No internal comparison with the selected time is possible in the interrogate sequence.)

3.3.1.10.6 Fuze option memory fault. The setter shall accept fuze option memory outputs if one, and only one, option memory output is present in each fuze channel, and these two single outputs agree with the option number selected during the test/set/verify sequence.

3.3.1.10.6.1 Fault detection in test/set/verify sequence. The setter shall detect an option fault during the test/set/verify sequence when either or both option memory channels has an output which differs from the one selected, has multiple option memory outputs detected in one or both channels, or has no option memory outputs detected in either or both channels. When an option fault is detected in the test/set/verify sequence the FUZE STATUS panel shall display FAIL.

3.3.1.10.6.2 Fault detection in interrogate sequence. In the interrogate sequence the setter shall detect an option fault when the channel option memory outputs are not identical, when multiple option memory outputs are detected in one or both channels, or when no option memory outputs are detected in either or both channels. When one or more of the above option faults are detected in the interrogate sequence, the FUZE STATUS panel shall display FAIL 8.

3.3.1.11 Induced fault safety requirements. Induced fault safety refers to the setter's ability to sustain induced faults in its own critical circuits and still preclude an operator's accepting an incorrectly set or faulty fuze which, if used, would affect inflight nuclear safety. See paragraph 6.5.

3.3.1.11.1 Fuze battery detection circuit fault. The fuze battery fault detection specification (3.3.1.10.1) shall apply with each redundant detection circuit successively disabled at the setter test connector J7.

3.3.1.11.2 Time 2/3 decode logic fault. The fuze accuracy fault detection specification (3.3.1.10.5) shall apply with each redundant 2-out-of-3 logic circuit in the setter successively disabled at the setter test connector J7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.1.11.3 Time accuracy error detection fault. When a simulated erroneous (nonselected) fuze time-out is applied to the setter during a test/set/verify sequence, and the time accuracy error detection circuit (3.3.1.10.5) is disabled at the setter test connector J7, the resulting time FUZE STATUS display on the setter shall be the nonselected time, or a display which conveys to the operator an obvious discrepancy between selection and displayed result.

3.3.1.11.4 Setter time base fault. When a setter time base fault produces a shift in frequency of greater than ± 8% the setter shall react to such a fault by detecting a false error in the fuze clock frequency test (3.3.1.10.3) and displaying FAIL in the test/set/verify sequence, and FAIL 6 in the interrogate sequence. Any other displayed result which conveys to the operator an obvious discrepancy between setting selection and FUZE STATUS display is also acceptable when inducing a setter time base fault.

3.3.1.11.5 Option error detection fault. When the simulated nonselected fuze option outputs occur identically in both fuze channels and the option error detection circuit has been disabled at the test connector J7, the resulting option displayed in the FUZE STATUS display shall convey an obvious incorrect (nonselected) output.

3.3.2 Fuze setting requirements. The following requirements apply to the setter when powered per 3.3.1.1.

3.3.2.1 Fuze setting. The setter shall meet the accuracy requirements specified below when used to set and interrogate the memories of accepted fuzes (MIL-F-48703) from current production. Each fuze shall be set, verified, and interrogated three times as follows: (1) 13.0-1, (2) 70.0-2, and (3) 125.0-3. A failure to meet any of the accurcy requirements during each setting of any fuze shall constitute a setter failure to meet this requirement.

Requirement	Requirement Paragraph
Time Setting accuracy	3.3.1.8.2.1
Option setting accuracy	3.3.1.8.2.2
Time verification accuracy	3.3.1.8.2.3
Option verificaiton accuracy	3.3.1.8.2.4
Interrogate time accuracy	3.3.1.8.3.1
Interrogate option accuracy	3.3.1.8.3.2

3.3.2.2 Fuze memory retention. The fuzes which were set in 3.3.2.1 shall be interrogated again by the same setter after 24 hours have elapsed from the time the fuzes were last set. The setter shall demonstrate that the fuze setting memory is maintained and displayed as specified in 3.3.1.8.3.1 and 3.3.1.8.3.2 following the 24 hour interval. The setter and fuze temperatures shall be maintained constant within ± 3°C during the interval.

3.3.3 Human factor requirements.

3.3.3.1 Keyboard contact force. Each key on the 12-key data select keyboard shall require a contact force of 16 ± 5 ounces to produce a setter data entry.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.3.2 Keyboard debouncing. Even pressure contact on any key shall result in a single digit entry on the SELECT display.

3.3.3.3 LED brightness. Each LED character in both display panels of the assembled setter shall have a minimum brightness of 0.05 foot Lamberts.

3.3.3.4 Night keyboard visibility. Each of the six keyboard illuminating LED's shall have a minimum brightness of 0.6 foot Lamberts.

3.3.3.5 Display functions.

3.3.3.5.1 Setter busy signal. One digit in the FUZE STATUS display shall be a blinking digit during the time interval between completion of keyboard data entries and fuze memory playback to indicate the setter's automatic test and setting sequences are being performed.

3.3.3.5.2 Status display retention. The FUZE STATUS display shall be retained at the completion of a test/set/verify or interrogate sequence until power is extinguished, or until a new select entry has been completed.

3.3.3.5.3 Time-option dash. The time and option sections of each display panel shall be separated by a LED dash segment. The dash in the SELECT display shall not appear until the time entry has been completed to indicate the next digit to be entered is the option entry.

3.3.3.5.4 Decimal point sequence. When power is applied to the setter, the SELECT display shall be a zero digit followed by a decimal point to indicate setter operational readiness and to establish for the operator the decimal digit positions. After the first digit entry, the decimal point shall be extinguished to allow its reappearance when keyed in prior to entry of the tenths of second digit.

3.3.3.6 Fuze insertion-removal. The force required to insert a fuze into the setter connector receptacle sufficient to establish electrical contact (3.3.1.4.4) shall not exceed 25 lbs. The force required to remove a fuze shall not exceed 25 lbs.

3.3.3.7 Setter weight. The weight of the complete setter assembly shall not exceed 22 pounds.

3.3.4 Power supply requirements.

3.3.4.1 Setter power control. Power shall be applied to the setter when the operator initially presses the CE key. Power shall be removed from the operating setter when the operator presses the CE key twice.

3.3.4.2 Automatic turn-off. Setter power shall be automatically turned off after 36 ± 5 seconds have elapsed from the time of the last CE, data, or sequence select keystroke entry.

3.3.4.3 Power cable requirements. The following specifications apply to the external power cable (11711399) used to provide external operating power to the setter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.4.3.1 Resistance. The resistance between connector contacts on each conductor shall be less than 0.35 ohms.

3.3.4.3.2 Current. Each conductor and associated connector contacts shall be capable of carrying 3A dc maximum, and 40A for 50 milliseconds.

3.3.4.3.3 Isolation. The isolation resistance between conductors shall be 10M ohms minimum.

3.3.4.4 Power indicator. The keyboard lamps shall be activated when the power cable is connected to an external power source.

3.3.4.5 EMR rejection filter. The low pass EMR filter in the external power input circuit shall have the following rejection versus frequency characteristic when installed in the setter with external power applied and the setter operating.

Frequency	Minimum attenuation
30KHz	15 dB
150 KHz to 100 MHz	25 dB

3.3.5 Environmental requirements. Unless otherwise specified, the complete setter assembly (including outer case and power cable) shall meet the requirements specified in 3.3.1 through 3.3.4 following subjection to the environmental condition specified below. When specified in the following paragraphs, the setter assembly shall also meet 3.3.1 through 3.3.4 when operated in the environment.

3.3.5.1 Air temperature extremes. The setter shall be operated at -34°C and +55°C in accordance with 4.6.5.1.

3.3.5.2 Temperature-humidity. The setter shall be subjected to the temperature and humidity environments described in 4.6.5.2.

3.3.5.3 Thermal shock. The setter shall be subjected to the thermal shock environment described in 4.6.5.3.

3.3.5.4 Pressure. The setter shall be subjected to the pressure environment described in 4.6.5.4.

3.3.5.5 Salt fog.

3.3.5.5.1 Salt fog. The setter shall be subjected to the salt fog environment described in 4.6.5.5.1. The case cover shall be opened for 1 minute prior to completion of the conditioning.

3.3.5.5.2 Salt fog (safety). The setter shall be subjected to the salt fog environment described in 4.6.5.5.2. The case cover shall be open, and the connector cover opened for 5 minutes prior to completion of the conditioning. The setter need not be operational following this conditioning, however it shall not produce a safety fault which would allow an operator to accept an incorrectly set fuze.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.5.6 Sand and dust.

3.3.5.6.1 Sand and dust. The setter shall be subjected to the sand and dust environments described in 4.6.5.6.1. One minute prior to completion of the conditioning, the case cover shall be opened.

3.3.5.6.2 Sand and dust (safety). The setter shall be subjected to the sand and dust environments described in 4.6.5.6.2. The case cover and connector cover shall be open for 5 minutes. The setter need not be operational following completion of this conditioning, however it shall not produce a safety fault which would allow an operator to accept an incorrectly set fuze.

3.3.5.7 Rain.

3.3.5.7.1 Rain. The setter shall be subjected to the rain environment described in 4.6.5.7.1. One minute prior to completion of the rain conditioning, the case cover shall be opened.

3.3.5.7.2 Rain (safety). The setter shall be subjected to the rain environment described in 4.6.5.7.2. The case cover shall be open during the rain conditioning, and the connector cover shall be opened 5 minutes prior to completion of the conditioning. The setter need not be operational following this conditioning, however, it shall not produce a safety fault which would allow an operator to accept an incorrectly set fuze.

3.3.5.8 Fungus. The setter shall be subjected to the fungus environment described in 4.6.5.8.

3.3.5.9 Seal. The setter shall be subjected to the seal environment described in 4.6.5.9.

3.3.5.10 Surface and air transportation vibration.

3.3.5.10.1 FAAS requirement. The setter shall be subjected to the vibration environments described in 4.6.5.10.1.

3.3.5.10.2 Production acceptance requirement. The setter shall be subjected to the modified vibration environment described in 4.6.5.10.2 (see 4.4.5.1).

3.3.5.11 Impact.

3.3.5.11.1 Side drop. The setter shall be subjected to the side drop environment described in 4.6.5.11.1.

3.3.5.11.2 Corner drop. The setter shall be subjected to the corner drop environment described in 4.6.5.11.2.

3.3.5.11.3 Edge drop. The setter shall be subjected to the edge drop environment described in 4.6.5.11.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

3.3.5.11.4 Production acceptance drop test. The setter shall be subjected to a modified drop environment described in 4.6.5.11.4 for production acceptance testing (see 4.4.5.1).

3.3.5.12 Ice fog. The setter shall be subjected to the ice fog environment described in 4.6.5.12.

3.3.5.13 EMR. The setter shall be operated during, and following subjection to the EMR environment described in 4.6.5.13.

3.3.5.14 Lightning EMP. The setter shall be operated during, and following subjection to, the lightning EMP environment described in 4.6.5.14.

3.3.5.15 Static electricity. The setter external connector pins, with the exception of E/M A(J3-S4), E/M B(J3-S5), E/M C(J3-S14), and BTRY MON (J3-S7), shall be subjected to the static electricity environment described in 4.6.5.15.

3.3.5.16 Transient radiation (nuclear). The setter shall be operated during, and following subjection to, the gamma dot transient radiation environment described in 4.6.5.16. The setter need not be operational during exposure to this environment, however it shall not produce a safety fault which would allow an operator to accept an incorrectly set fuze. The setter shall be fully operational as specified in 3.3.2 following subjection to this environment.

3.3.5.17 EMP (nuclear). The setter shall be operated during, and following subjection to, the nuclear EMP pulse environment described in 4.6.5.17. The setter need not be operational during exposure to this environment, however it shall not produce a safety fault which would allow an operator to accept an incorrectly set fuze. The setter shall be fully operational as specified in 3.3.2 following subjection to this environment.

3.3.5.18 Life. The setter shall be subjected to 2000 powered set and interrogate cycles as described in 4.6.5.18. See section 6.4 for design objectives for setter storage life.

3.4 Workmanship. All parts shall be manufactured and finished in a thoroughly workmanlike manner to insure satisfactory functioning and durability. (See MIL-A-2550, Condition of Materials, Parts and Assemblies.)

3.4.1 Soldering. Soldering shall be in accordance with drawing 11711877.

3.4.2 Printed circuit boards. The printed circuit wiring boards shall meet the acceptance limits specified in MIL-P-55110 and MIL-STD-275 as modified by the individual board drawing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.1.1 Contract quality assurance system. The contractor shall provide and maintain a quality assurance system in compliance with MIL-Q-50829.

4.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification in compliance with MIL-Q-50829.

4.3 First article approval sample. The first article approval sample as specified in 3.1.1 shall be examined and tested in accordance with the First Article Approval Sample tests (4.4.5.2), Classification of Defects (4.4.4), and Material and Component Certification (4.4.6). The tests shall be performed by the contractor under the observation of the Government QA representatives and the representative of the cognizant technical agency. The contractor shall supply complete data in reproducible form on the contractor tests and examinations of the first article approval sample units. The data and tested samples shall be delivered to the technical agency.

4.3.1 Approval of first article sample. If the sample passes the criteria of specified examinations and tests, it will be approved. If the sample fails any of the examinations and tests, a failure analysis shall be conducted by the contractor under Government surveillance. The results of the tests and failure analysis of the units, together with the engineering analysis of the units, shall form a basis for corrective action. Depending upon the degree of corrective action deemed necessary by the Government, the first article approval sample may be:

a. Conditionally approved with the provisions that the recommended changes are made by the contractor prior to start of regular production or during production of the first lot as prescribed by the Government. Such portions of first article inspections needed to verify corrective action may be included in the acceptance requirements of the lot, at the Government's discretion.

b. Disapproved, and new samples required for the tests failed and for related tests which might be invalidated by the corrective action.

c. Disapproved, and a complete new sample required for approval. In all cases, the contractor shall comply with any required changes for the duration of the contract.

4.3.1.1 Reinstitution of tests. Acceptance of the first article approval sample does not relieve the contractor from meeting all requirements of this specification throughout the contract. The Government reserves the right to independently verify all requirements by repeating first article approval sample tests or any portion thereof of any production lot. If such verification indicates failure to meet requirements previously verified by first article appproval sample tests, the contractor will be required to take corrective action. If the requirements are not met on the first setter to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

which the correction can be applied, the Government reserves the right to institute tightened inspection by adding to the normal lot acceptance inspections all or such parts of the first article approval sample tests needed to verify corrective action. Normal acceptance inspection will be resumed after two consecutive setters have satisfactory passed the tightened conditions.

4.4 Acceptance inspection. Inspection shall be as specified in MIL-A-2550 and in this document. The contractor shall notify the Government QA representative prior to proceeding with inspection for Classification of Defects (4.4.4) and Acceptance Tests (4.4.5). Inspection may be made at a higher assembly level provided that equal quality assurance provisions are satisfied, but not at a lower level than specified.

4.4.1 Sampling. Every setter shall be tested as required in Section 4.4.2.

4.4.2 Setter production acceptance. Each setter assembly shall be accepted on the basis of Classification of Defects (4.4.4), Acceptance Tests (4.4.5) and Material and Component Certification (4.4.6).

4.4.2.1 Disposition of rejects. Rejected M38 fuze setter assemblies may be resubmitted with the concurrence of the procuring agency in accordance with the following:

a. Resubmitted setters shall have been reworked to repair all discrepancies disclosed during previous submission to the tests.

b. Rework shall follow the same techniques and specification used in manufacturing the setter and shall be approved by the designated technical agency.

c. The resubmitted setter shall be resubjected to the test in which the failure occurred and any other test required by the designated technical agency to demonstrate acceptability.

d. When the resubmitted setter passes the tests in step c above, then the normal test sequence shall be resumed.

e. No setter may be resubmitted if the unit fails the retest after repairs are effected by replacing the same part twice or making the same adjustment twice without concurrence from the designated technical agency.

f. No setter may be resubmitted after it has been rejected by the same test five times without concurrence from the designated technical agency.

g. No setter may be submitted to envirnomental testing more than once without concurrence from the designated technical agency.

4.4.3 Data recording. A written record shall be made of the results of all examinations and tests performed as specified elsewhere in section 4 of this specification. Data from each operating test shall be recorded, but only the defect summary need be recorded in the inspection of the classification of defects. If any inspections are conducted on a subassembly basis, the acceptance data for each lower assembly grouping shall be so identified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.4.3.1 Records of production acceptance tests. Production acceptance test data on each setter assembly shall be assembled into an approved log data book in duplicate and identified by the assembly serial number. After Government acceptance of the assembly, the duplicate copy of the data book shall be shipped as directed by the procuring activity. The original data book shall be retained by the contractor for the duration of the contract. This book shall include the following sections:

a. Log Contents and end item contractor certification;
b. Subassembly serial number identification;
c. Major parameter data summary;
d. System data analysis (brief overview of all system test data noting test failures and repairs prior to acceptance).
e. Setter system electrical test data records.
f. Mechanical final assembly data record, including data from environmental tests required for production acceptance.
g. Applicable Engineering Change Proposals (ECP's), Request for Waivers (RFW's), and Request for Deviations (RFD's).
h. Any other sections required to compile the assembly history.

4.4.4 Classification of Defects. The assembly shall be inspected by using the Classification of Defects and applicable drawings.

4.4.4.1 Parts and assemblies.

M38 Fuze Setter Classification of Defects

Categories	Defects	Acceptance Criteria	Method of Inspection
ADAPTER ASSEMBLY (DWG 11735216)			
Critical	None defined		
Major			
101	Damaged connector - 6 places	100 Percent screening	DS-1
102	Lid spring force less than 50 gms applied to tab (to lift lid off adapter)	100 Percent screening	Gauge
103	Connectors sealed all around - 2 places	100 Percent screening	Visual
Minor	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

FENCE/SHIELD ASSEMBLY (DWG 11735218)

Critical	None defined		
Major			
101	Staking in place, resistors and LED's	100 Percent screening	Visual
Minor	None defined		

CHASSIS ASSEMBLY (DWG 11735215)

Critical	None defined		
Major			
101	Switching unit damaged	100 Percent screening	Visual
102	Fence gasket in place	100 Percent screening	Visual
103	Feed throughs on chassis panel	100 Percent screening	Visual
104	EMR screen over display	100 Percent screening	Visual
105	Adapter gasket in place	100 Percent screening	Visual
Minor	None defined		

OUTER CASE ASSEMBLY (DWG 11735205)

Critical	None defined		
Major			
101	Pressure relief valve properly installed	100 Percent screening	Visual
Categories	Defects	Acceptance Criteria	Method of Inspection
102	Latches and hinges functional	100 Percent screening	Visual
Minor			
203	Marking illegible or missing	100 Percent screening	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

M38 FUZE SETTER (DWG 11727881)

Critical	None defined		
Major			
101	LED display dot segments functional and completely visible (4 x 7 dot characters, 13 places and decimal points, 2 places)	100 Percent screening	Visual
102	Keyboard side lamps functional - 6 places	100 Percent screening	Visual
103	EMR gasket in place	100 Percent screening	Visual
104	⌖ A B C .005 2-21S connector	100 Percent screening	Gage/fixture
105	1.154 min/1.170 max seat-to-connector face depth - 2 places	100 Percent screening	Gage/fixture
106	Damaged interface connectors - 2 places	100 Percent screening	Visual
107	Power connector damage J1-S	100 Percent screening	DS-1
108	Verify proper wiring and placement of leads between connectors and chassis/case assembly	100 Percent screening	DS-1
109	$0.136^{+0.001}_{-0.000}$ x 0.39 MIN DP - 3 holes	100 Percent screening	Gage/fixture
110	⌖ A B Ø.005 3 places, hole locations	100 Percent screening	Gage/fixture
111	1.449 min inside depth	100 Percent screening	SME

Categories	Defects	Acceptance Criteria	Method of Inspection
112	0.405 min seat-to-top of flange	100 Percent screening	SME
113	2.570 dia. min internal diameter	100 Percent screening	SME
114	3.045 dia. max seat external diameter	100 Percent	SME

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

114	3.045 dia. max seat external diameter	100 Percent screening	SME
115	Verify proper orientation of connector 2 places	100 Percent screening	Visual
116	Protection cap gasket in place	100 Percent screening	Visual
117	Power cable damage	100 Percent screening	DS-1, Visual
Minor			
201	Power cable secured by tiedowns	100 Percent screening	Visual
202	Scratched display window - 2 places	100 Percent screening	Visual
203	Marking illegible or missing	100 Percent screening	Visual

4.4.4.2 Defect standards.

DS-1 - Any damage to connector pins, sockets or housing such that proper assembly of mating connector is hampered or prohibited shall consitute a defect.

4.4.5 Acceptance tests. All acceptance tests, except as noted, shall be performed by the contractor in accordance with the following paragraphs. Failure reports shall be submitted to the cognizant technical agency for any test failed in production acceptance, first article approval acceptance, and comparison sample tests.

4.4.5.1 Production acceptance tests. The complete setter assembly shall be subjected to the following environmental acceptance tests in the order listed and then subjected in any order to the electrical acceptance tests listed thereafter. All acceptance tests shall be performed on each setter at room ambient temperature; in addition tests specified in 4.6.1 (Functional requirements), except 4.6.1.9.8, shall be performed at extreme cold and hot temperatures in accordance with 4.6.5.1.

Environmental Requirement	Requirement Paragaraph	Test Paragraph
Vibration	3.3.5.10.2	4.6.5.10.2
Impact	3.3.5.11.4	4.6.5.11.4
Seal	3.3.5.9	4.6.5.9
Static electricity	3.3.5.15	4.6.5.15

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Electrical Requirement		
Functional requirements	3.3.1	4.6.1 (except 4.6.1.1.1.2, 4.6.1.2.1.2, 4.6.1.2.2.2, 4.6.1.2.3.2 and 4.6.1.3.2)
Fuze setting requirements	3.3.2	4.6.2 (except 4.6.2.1.2 and 4.6.2.2.1)
Display functions	3.3.3.5	4.6.3.5
Power supply requirements	3.3.4 (except 3.3.4.3.2)	4.6.4 (except 4.6.4.3.2)

4.4.5.2 First Article Approval Sample Inspection. The first article approval sample shall be subjected to the production acceptance requirements (4.4.2) and the following additional first article tests performed at the temperature(s) indicated and in the group sequence listed. Tests within each group may be performed in any order. Acceptance criteria for the sample test is zero rejects for each group.

Group A, Electrical Requirements	Requirement Paragraph	Test Paragraph
Vehicle Power Supply (hot, cold, room)	3.3.1.1	4.6.1.1.1.2
Vehicle Power noise immunity (hot, cold, room)	3.3.1.2.	4.6.1.2.1.2 4.6.1.2.2.2 4.6.1.2.3.2
G76 Noise Immunity	3.3.1.3	4.6.1.3.2
Fuze Setting (hot,cold)	3.3.2	4.6.2
Fuze Memory Retention (hot,cold,room)	3.3.2.2	4.6.2.2.1
Human Factors (room)	3.3.3	4.6.3
Power Cable Current (room)	3.3.4.3.2	4.6.4.3.2
Group B, Environmental Requirements		
Temperature-humidity	3.3.5.2	4.6.5.2
Thermal Shock	3.3.5.3	4.6.5.3
Pressure	3.3.5.4	4.6.5.4
Salt Fog	3.3.5.5.1	4.6.5.5.1
Sand and Dust	3.3.5.6.1	4.6.5.6.1
Rain	3.3.5.7.1	4.6.5.7.1
Fungus (note 1)	3.3.5.8	4.6.5.8
Vibration (hot,cold)	3.3.5.10.1	4.6.5.10.1
Impact (hot,cold)	3.3.5.11 (except 3.3.5.11.4)	4.6.5.11 (except 4.6.5.11.4)
Ice Fog	3.3.5.12	4.6.5.12
EMR (note 1)	3.3.5.13	4.6.5.13
Lightning EMP (note 1)	3.3.5.14	4.6.5.14
EMP (note 1)	3.3.5.17	4.6.5.17
Life	3.3.5.18	4.6.5.18

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Group C, Post Environment Electrical Requirements

Function Requirements (room)	3.3.1	4.6.1 (except 4.6.1.1.1.2 4.6.1.2.1.2 4.6.1.2.2.2. 4.6.1.2.3.2 4.6.1.3.2)

Group D, Environmental Safety Requirements (Note 1)

Salt Fog	3.3.5.5.2	4.6.5.5.2
Sand and Dust	3.3.5.6.2	4.6.5.6.2
Rain	3.3.5.7.2	4.6.5.7.2

Note 1 - These environmental tests will be conducted by the Government Technical Agency, or may be waived at the discretion of the Government Technical Agency. Performance tests on the setter following subjection to these environments shall be conducted by the contractor. First article approval shall be based on the test requirements of 4.4.5.2 and only those tests to be conducted by the Government Technical Agency specifically required by the contract (see paragraph 6.1.6).

4.4.5.3 Comparison sample tests. The comparison sample setters shall be selected at random as specified in the applicable contract by the resident inspector from setters which were successfully subjected to the setter acceptance tests listed in 4.4.5.1. These tests will be observed by the resident Government inspector and a representative of the designated technical agency. The comparison sample test sequence shall be any combination of, or all of the FAAS tests as specified by the Government technical agency. The contractor shall supply complete data in reproducible form on the contractor tests of the comparison samples. All data and samples shall be forwarded to the designated technical agency who may conduct other tests and engineering analyses. All comparison sample tests shall be performed by the contractor and witnessed by the resident Government inspector, and a representative designated by the procuring agency.

4.4.5.3.1 Comparison sample failures. If an assembly fails any of these tests, testing shall be stopped. The contractor shall analyze the failure and forward data and recommendations to the technical agency within five working days. The technical agency will review the analysis and recommendations, and direct changes and retests where indicated.

4.4.6 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certification or statement of findings, that only materials and components conforming to the item specification and drawings have been used.

4.5 Test conditions and equipment.

4.5.1 Test conditions. Unless otherwise specified, all measurements are to be made at normal ambient room conditions of temperature, relative humidity and altitude. Temperature conversion between Fahrenheit and Celsius shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

carried to the nearest whole degree. The following table of tolerances shall be applied where applicable to specified test conditions unless otherwise defined in the test procedures:

a.	Temperature in test chamber	± 2°C
b.	Relative humidity	± 5 percent
c.	Vibration amplitude	± 10 percent
d.	Electrical and physical measurements except event time	± 3 percent
e.	Event time	± 2 milliseconds
f.	Time base for fuze clock test	± 50 ppm

4.5.2 Inspection equipment. Inspection equipment shall be in accordance with the design level shown by the applicable EDL and of sufficient accuracy and quality to permit performance of the required inspection. Gage and instrument setting shall be a function of the equipment, and independent of the operator. Equipment shall be marked to indicate the limit of acceptance for each test. Test procedures and calibration procedures shall conform to MIL-Q-50829.

4.5.3 Simulated fuze loads. The setter interface outputs shall be loaded by the simulated fuze impedance levels specified below except when setter tests are conducted with an actual fuze.

Fuze Function	Fuze Connector/Pin Nos.	Load Z (±5%)
Erase/Memorize	J3-N/4-5-14	See 4.5.5.9
Set Control	J2-N/4-8-14	27K
Setter Clock	J3-N/6-15-16	27K
MNOS (option set clock)	J3-N/2-3-13	71K

4.5.4 External power source. The external power source for the M38 setter is a vehicular supply as defined by MIL-STD-1275 or a G76 generator as specified by MIL-G-49369. For purposes of testing, the external power voltage shall be 18.0 ± 0.2V except as noted in the tests listed below.

Paragraph Title	Paragraph No.
Input voltage	4.6.1.1.1
Input voltage regulation	4.6.1.1.3
Low voltage indicator	4.6.1.1.4
Power shutdown threshold	4.6.1.1.5
5 volt regulation	4.6.1.1.6
Vehicle power noise immunity	4.6.1.2
G76 noise immunity	4.6.1.3
Error display priority	4.6.1.7
Fuze connector resistance	4.6.1.9.8

4.5.5 Simulated fuze. All references to a simulated fuze or simulated fuze inputs shall be characterized by the following paragraphs. The simulated fuze must interface with only J2-S and J3-S connectors (specified in the fuze/setter interface drawing 11710170) in the adapter assembly. The simulated fuze shall be capable of providing the normal as well as the abnormal conditions required in 4.6.1 testing of a setter. Signal timing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

herein refers to setter-sequence step S13 (verification portion of the test/set/verify sequence) or S9 (set portion of the test/set/verify sequence).

4.5.5.1 Operating current. The simulated fuze current shall be 60 ± 5ma.

4.5.5.2 Continuity loop. Between J2-S pin 1 and J3-S pin 21 a resistance of no more than 0.1 ohm shall be installed. Under setter fault testing conditions this resistance shall be capable of being increased to 2.0 ohms ±1% or 5.0 ohms ± 1%. These resistors shall have a minimum power rating of 3.0 watts.

4.5.5.3 MNOS Enable. MNOS Enable (J2-S pins 2, 3 and 13) is a bidirectional signal with various uses. During the majority of the setter operations, the setter holds MNOS Enable A and B at a high logic state. During the option prejam (3.3.1.8.1.2) these two MNOS Enable lines are clocked at 1.28 kHz. During S13 (setter verification time) of the setter's sequencing, all Setter Control of these lines is removed and the lines are monitored by the setter. To simulate normal fuze operation, a logic high signal shall be supplied when a valid timeout minus 120 counts should occur. This logic high signal shall be sourced from a 56K ohm impedance source until loss of the setter's Setter Cont signal has been detected. Upon the loss of Setter Cont the source impedance shall be reduced to a minimum of 150 ohms.

4.5.5.4 Setter control. The simulated fuze shall load Setter Control A, B and C (J2-S pins 4, 14 and 8 respectively) with a 27K ohm ± 10% resistor to ground.

4.5.5.5 Option sets. Option set outputs to the setter shall be provided upon detection of the verification sequence (S13) of the setter and they shall be maintained until Fuze Power is removed. The normal option set outputs shall be provided as determined by the option clock count prejam (4.5.5.3). During testing of the setter fault detection circuits, all option set lines shall be capable of either being a logic high or a logic low.

4.5.5.5.1 01A,01B. The simulated fuze shall supply an 01A and 01B (J2-S pins 9 and 19) signal at the appropriate time if the option clock count prejam is 226.

4.5.5.5.2 02A, 02B. The simulated fuze shall supply an 02A and 02B (J2-S pins 7 and 17) signal at the appropriate time if the option clock count prejam is 258.

4.5.5.5.3 03A, 03B. The simulated fuze shall supply an 03A and 03B (J2-S pins 5 and 15) signal at the appropriate time if the option clock count prejam is 292.

4.5.5.6 ± 13V supply. The simulated fuze shall provide the setter with 13.7 ± 0.1V dc (J2-S pins 10, 11, and 20) whenever Fuze power supplied by the setter is detected. Each of three 13V lines shall be capable of supplying a minimum of 20 mA.

4.5.5.7 Decode CLK. A Decode CLK signal (J2-S pins 12, 18 and 20) shall be provided to the setter with a nominal period of 100 ms. This signal shall have a logic high for 25 ms and low for 75 ms. This signal shall be variable to periods of 92 ms, 94 ms, 106 ms and 108 ms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.5.5.8 Fuze oscillator input signal. The oscillator output signals from each fuze channel (J3-S Pins 1, 8 and 11) are rectangular pulses, 50% ± 10% duty cycle having a high logic level of 13.4 ± 1.5V and a low logic level of 0 + 1V. The frequency of each of these oscillators is 10240 Hz ± 3%. These oscillator inputs to the setter are used by the setter to determine the resultant setter clock frequencies during the setting and verifying portions of the test/set/verify sequence.

4.5.5.9 Erase/Memorize. The Erase/Memorize lines address the MNOS memory bias network in the fuze. This bias network consists of a 20.5V dc regulated source, a diode and 22K ohm source impedance, and a 100K ohm load. All setter tests with a simulated fuze shall have this circuit applied to the setter E/M lines (J3-5 pins 4,5, and 14). The 20.5V supply shall be activated whenever fuze power is applied from setter to fuze.

4.5.5.10 Setter Clock. The Setter Clock signals shall be loaded as specified in 4.5.3. In addition these lines shall be used to determine the number of counts each channel receives during the set portion (S9) of the test/set/verify sequence (4.6.1.8.2). These counts shall be directed to an up/down counter which is shifted from the up mode to down mode before the verification portion of the sequence is initiated. Again the Setter Clock shall be used to clock this counter down. When the counter reaches the all logic low state a MNOS Enable signal shall be generated. While generating fault conditions the up/down counter shall be capable of being altered to achieve the desired results.

4.5.5.11 Battery monitor. The Battery Mon line shall nominally be loaded by a 1.0 megohm ± 5% resistor to ground. In addition, when simulating fuze faults this impedance shall be variable to 500K ohms and 1.5 megohms.

4.6 Test procedures. The following paragraphs are intended tests, however, alternate tests may be used on approval of the procuring agency.

4.6.1 Functional tests. These tests are performed with the chassis assembly (11735215). Connections shall be made to:

1) J2-S and J3-S (adapter assembly)
2) J7 test connector
3) J6 power test connector
4) J8 external power connector

4.6.1.1 Power supply. The following conditions apply with power to the setter provided at the external connector J8.

4.6.1.1.1 Input voltage.

4.6.1.1.1.1 Production acceptance. Perform all testing per 4.6.1 using external power as specified in 4.5.4. Repeat testing of 4.6.1.8.2 with the exception of 4.6.1.8.2.5 with external power set to 33 volts.

4.6.1.1.1.2 FAAS acceptance. Perform all testing per 4.6.1 using external power as specified in 4.5.4. Repeat with external power set to 33 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.1.2 Input current. Maximum input current drain on the external power source shall be measured with the setter on and sequencing from S1 to S16.

4.6.1.1.3 Input voltage regulation. Measure the voltage at J6 with the setter power on and no keyboard data entry applied. Also, measure this voltage with +33 Vdc at J8.

4.6.1.1.4 Low voltage indicator. The external power source shall be decremented, in steps of 0.05 Vdc, from 15.0 Vdc. When the "B" is detected on the test connector J7 and the "B" indicator on the SELECT display occur, measure the voltage at the power test connector J6. This voltage shall be compared to the limits of section 3 for a pass/fail indication. Increase the external power source to 18.0 Vdc and verify that the "B" is still present at the test connector J7 and that the keyboard is inhibited. Clear the setter with a "CE" key entry and initiate a Test/Set/Verify sequence. When S13 of the setter sequence occurs, reduce the external power source to the level that caused the "B" indication to occur. Verify that the SELECT display is "0." and the STATUS display is halted and the "B" is detected at J7.

4.6.1.1.5 Power shutdown threshold. The external power source shall be decremented in steps of 0.05 Vdc from the external voltage level that "B" was detected (4.6.1.1.4). Measure the voltage at the power test connector J6 and monitor J7 for the F control voltage as the external power source is decremented. F control voltage level will drop to 0 volts upon reaching the no-operate threshold. The last voltage measured at the power test connector before F control dropped to 0 volts will be compared with the limits of section 3 for pass/fail.

4.6.1.1.6 5 volt regulator Measure the voltage level at F Control on the test connector while the setter is perfoming a Test/Set/Verify sequence and compare the minimum and maximum voltage measured with section 3 limits. This test shall be run with the input voltage at J8 adjusted to the "B" threshold level but with the "B" signal disabled at J7. The test shall be repeated with the input voltage at J8 set to 33 volts.

4.6.1.2 Vehicle power noise immunity.

4.6.1.2.1 Ripple. Ripple testing is accomplished by powering the setter with external power as described in figure 1.

4.6.1.2.1.1 Production acceptance Perform testing of 4.6.1.8.2 with the exception of 4.6.1.8.2.5 while subjecting the setter to the external ripple environment.

4.6.1.2.1.2 FAAS acceptance. Perform all testing as described in 4.6.1 with the exception of 4.6.1.2 while subjecting the setter to the external ripple environment.

4.6.1.2.2 Spikes. Spike testing shall be accomplished by externally powering the setter with power as described in figure 2.

4.6.1.2.2.1 Production acceptance. Perform testing of 4.6.1.8.2 with the exception of 4.6.1.8.2.5 while applying the external spike environment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

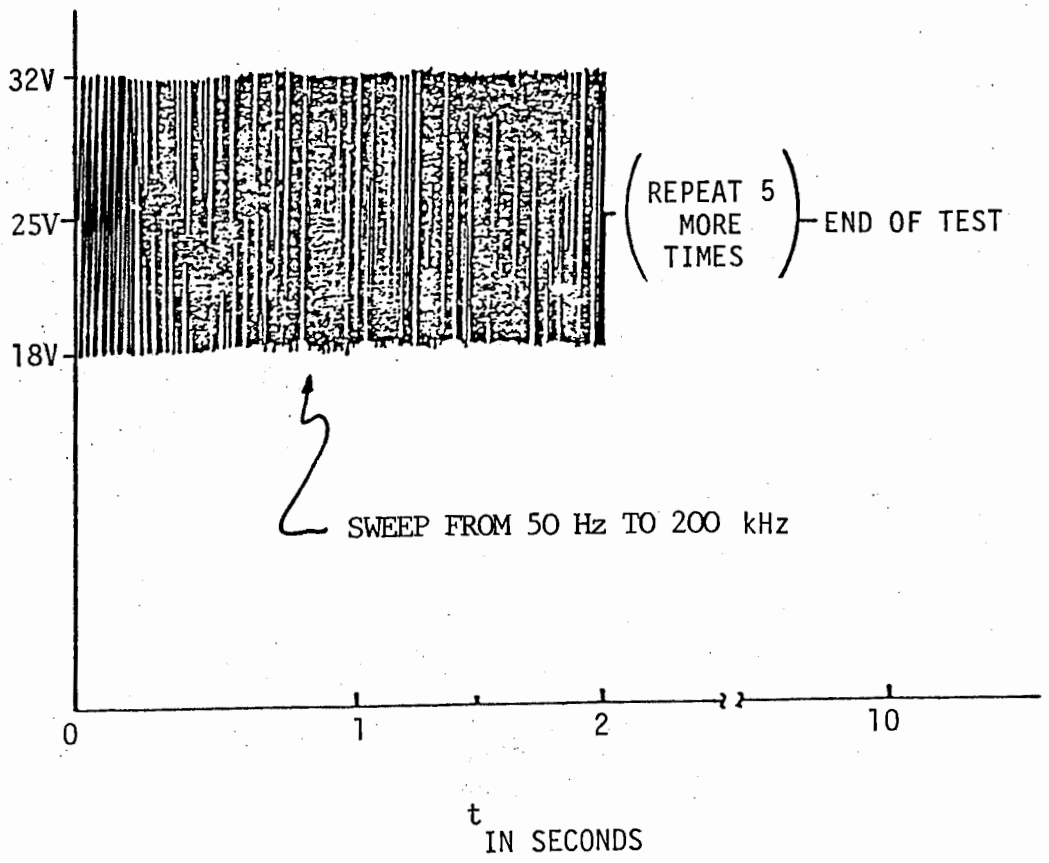


FIGURE 1. POWER SUPPLY RIPPLE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

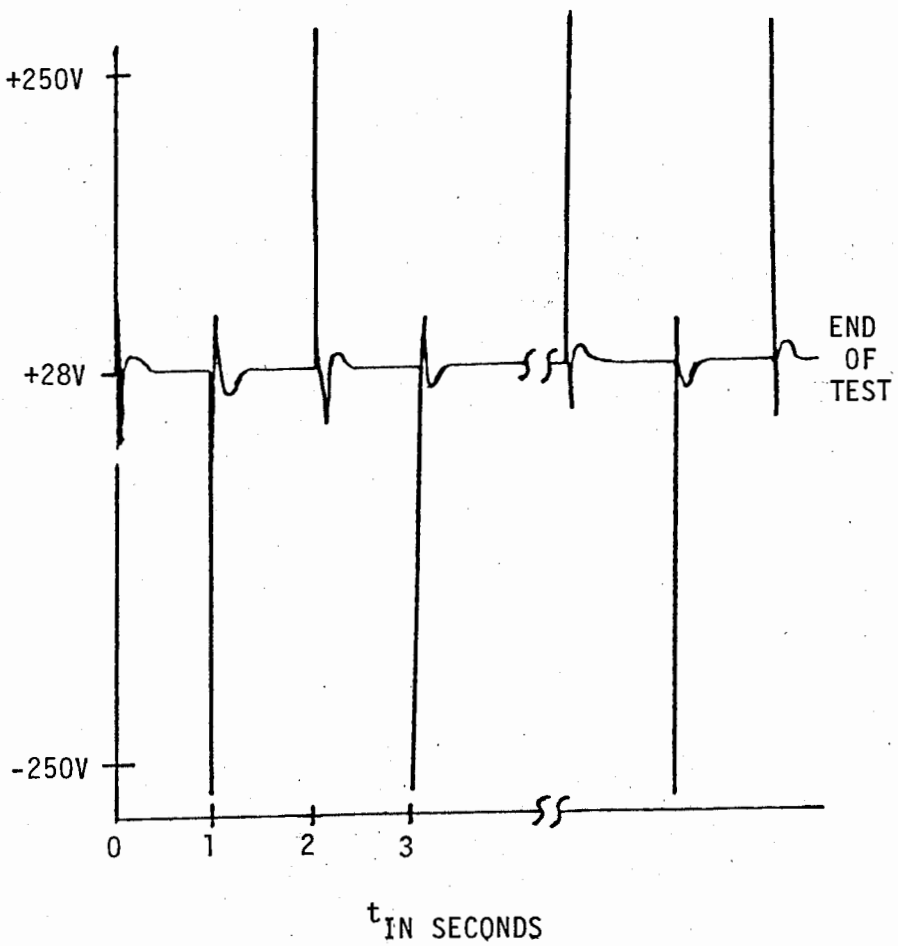


FIGURE 2. POWER SUPPLY SPIKES

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.2.2.2 FAAS acceptance. Perform all testing as described in 4.6.1 with the exception of 4.6.1.2 while applying the external spike environment.

4.6.1.2.3 Surges. Surge testing shall be accomplished while externally powering the setter with power as described in figure 3 (no fault condition) and figure 4 (single fault condition).

4.6.1.2.3.1 Production acceptance. Surge testing shall be accomplished while externally powering the setter with power as described in figure 3 and 4. Perform testing 4.6.1.8.2 with the exception of 4.6.1.8.2.5 while applying the external surges environment. Observe compliance with paragraph 3.3.1.2.3.1 (no fault condition) and paragraph 3.3.1.2.3.2 (single fault condition).

4.6.1.2.3.2 FAAS acceptance. Perform all testing of 4.6.1 with the exception of 4.6.1.2 while applying the external surge environment (figure 3 and 4). Observe compliance with paragraphs 3.3.1.2.3.1 and 3.3.1.2.3.2.

4.6.1.2.4 Vehicle starting disturbance. Input vehicle disturbance per figure 5 and repeat testing of 4.6.1.8.2 with the exception of 4.6.1.8.2.5 first with an option set 01 and then again with option set 02. Observe compliance with paragraph 3.3.1.2.4.

4.6.1.3 G76 noise immunity.

4.6.1.3.1 Production acceptance. Apply G76 generator power in the following manner. The G76 generator speed shall be adjusted such that the following conditions are met. First, the setter should turn on without a "B" indication. Second, a set/test/verify sequence will cause a "B" indication and shutdown does not occur. Third, with B at J7 grounded to disable the low voltage indication circuit, perform testing of 4.6.1.1.6 and 4.6.1.8.2 with the exception of 4.6.1.8.2.5.

4.6.1.3.2 FAAS acceptance. Perform all testing of 4.6.1 with the exception of 4.6.1.1.2, 4.6.1.1.4, 4.6.1.1.5, and 4.6.1.2 while applying power as specified in 3.3.1.3.

4.6.1.4 Setter self tests. In the following tests, when an error display on the setter is to be verified, a visual check shall be required on the first occurrence. Thereafter, only the error need be detected on the test connector unless otherwise specified. Similarly the detection of a sequence inhibit shall be made by observing the setter busy digit (4.6.3.5.1) for no indications on the first occurrence. Thereafter, a logic low on SEQ of the test connector may be used to determine a sequence inhibit.

4.6.1.4.1 169 prejam generator. The F Control input to the setter test connector J7-12 shall be held at a low logic level prior to powering a setter (4.6.4.1). An F error should occur and be present on the display and on the test connector J7. The sequence inhibit shall be determined by the lack of a setter busy digit and a low logic level on the SEQ line of the test connector.

4.6.1.4.2 Decode test counter. Testing for an F error indication shall be performed by applying a high logic level to the Decade Disable of the test connector while powering the setter (4.6.4.1). Observing the F error on the test connector. The SEQ line of the test connector shall also remain low.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

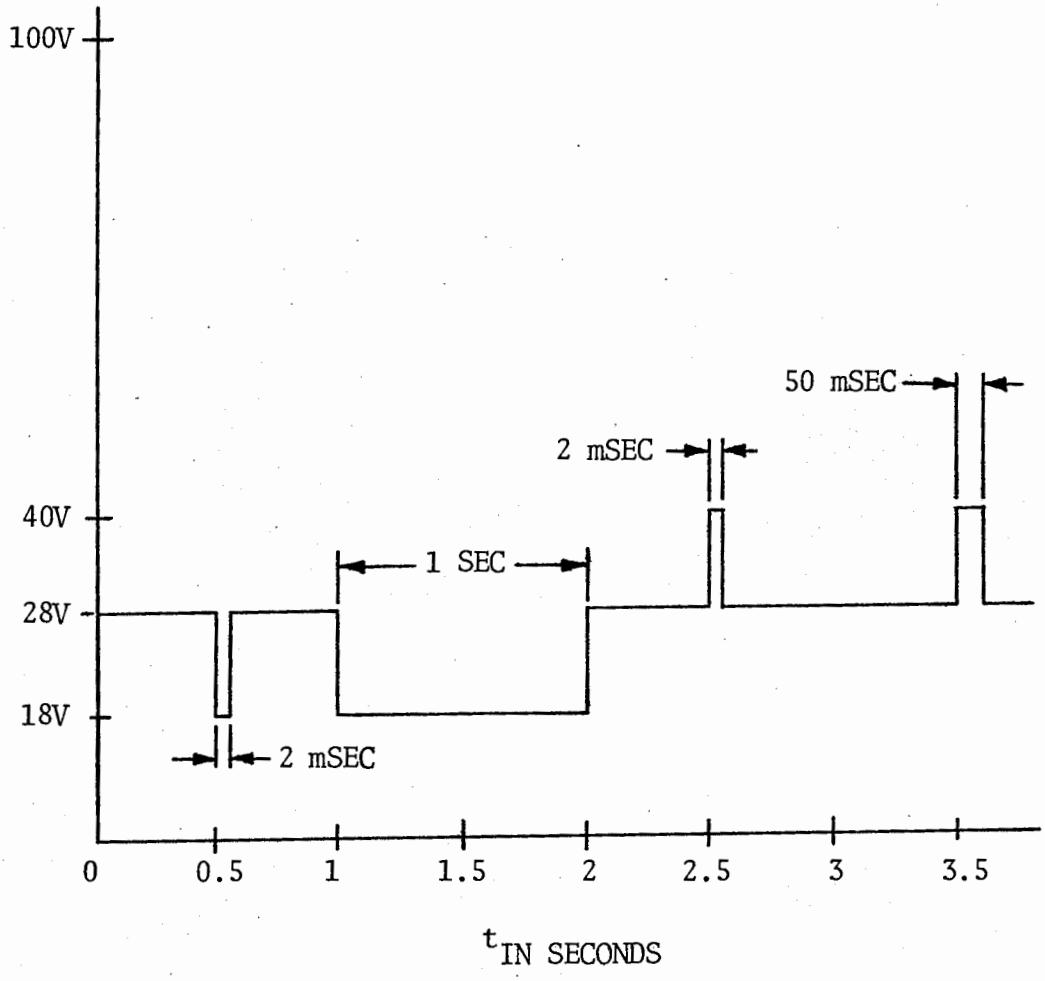


FIGURE 3. NO FAULT SURGE TESTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

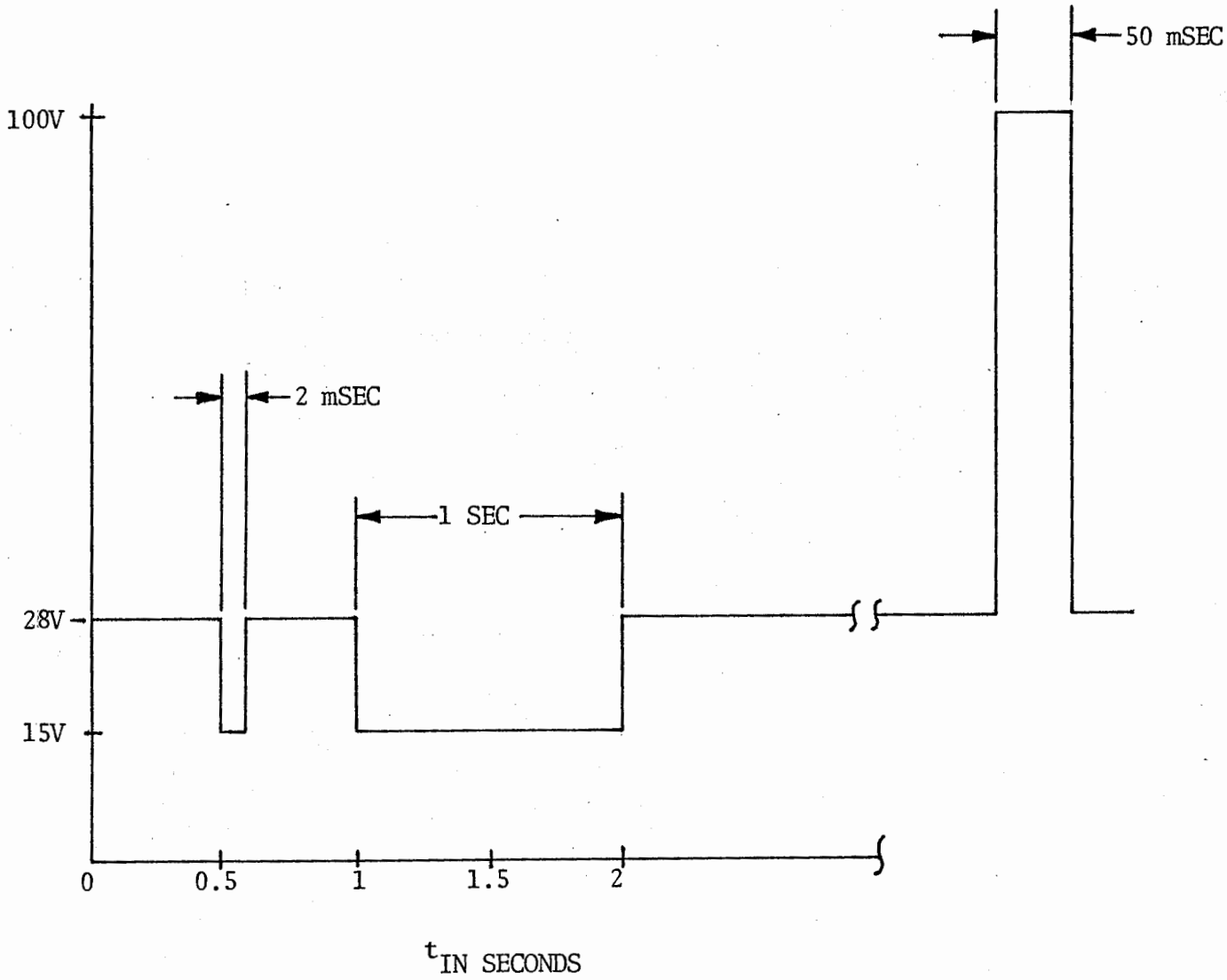


FIGURE 4. SINGLE FAULT SURGE TESTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

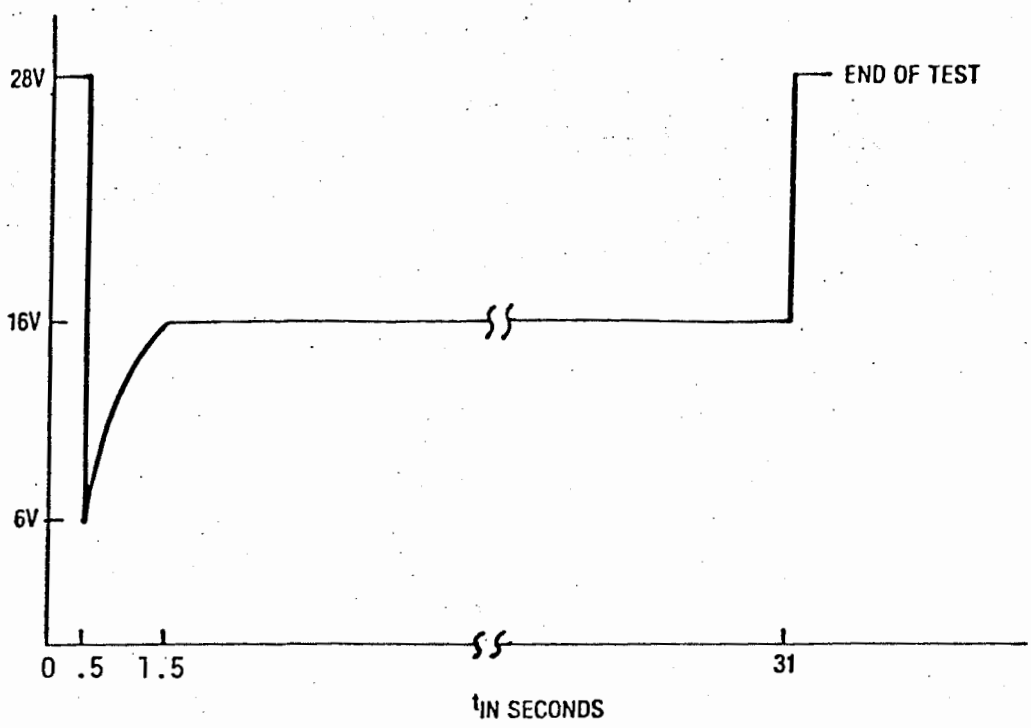


FIGURE 5. VEHICLE STARTING DISTURBANCES

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.4.3 Erase voltage output. All measurements are made on Erase/Memorize A, B, and C on J3-S.

4.6.1.4.3.1 Amplitude. A valid time and option entry per 3.3.1.5 shall be entered via J7. The Erase/Memorize lines shall each be loaded such that the erase voltage will be decreased to 25.0 ± 0.1V dc. Each Erase/Memorize line when loaded shall produce a logic high signal on F of the test connector when the other two E/M lines are not loaded. All Erase/Memorize lines shall be loaded to 27.0 ± .1V. This shall not produce a logic high on F of the test connector.

4.6.1.4.3.2 Error detection circuit. An F error on the test connector J7 shall be observed when a valid time and option entry 3.3.1.5 has been initiated via J7 with the Erase Voltage Disable on the test connector grounded. The setter sequence inhibited shall be determined by the absence of Fuze Power 1.5 seconds after a logic high on SEQ of the test connector has been detected.

4.6.1.4.4 Fuze/setter continuity.

4.6.1.4.4.1 Continuity fault detection. While powering the setter 4.6.4.1, simulated continuity loop resistance applied between J3-S pin 21 and J2-S pin 1 shall be increased from nominal to 2 ohms in accordance with 4.5.5.2. The setter shall not display a C when "1" key entry is made via J7. With a continuity resistance of 5 ohms, observe a C in the SELECT display of the setter upon making a numeric entry and determine that C on J7 is a logic high.

4.6.1.4.4.2 Power disable. During testing at 4.6.1.4.4.1 monitor Fuze Power J3-S. This voltage shall be no greater than 1.0V dc.

4.6.1.5 Keyboard entry/display. The following time entries, between 13.0 to 200.0, shall be entered on the setter and verified both on the SELECT display and by a logic high on SEQ of the test connector. SEQ indicates acceptance of the entry and automatic start of the setter sequencing. Verify that each entry is not removed until a CE is generated via J7 or the keyboard.

Enter 13.0 option 1
Enter 68.9 option 2
Enter 145.7 option 3
Enter 200.0 option 1

4.6.1.6 Entry error detection.

4.6.1.6.1 Time error. Time entries of 0.1, 12.9, and 200.1 shall be entered. The E shall be displayed in the SELECT display of the setter display and E on the test connect or J7 shall indicate the time entry error. Each error shall be removed upon a CE entry via J7.

4.6.1.6.2 Option error. Invalid option entries (0 and 4 through 9) shall be entered into the setter via J7 to verify that an E error on the test connector is detected. The SEQ line on J7 shall be a logic low 0.2 sec after the invalid option entry, thus indicating a sequence inhibit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.7 Error display priority. The setter prioritizes the setter errors. The following test sequences start with all errors present and then each error is removed one at a time until only the "E" error remains.

a. Set the external supply voltage to the level that "B" was detected (4.6.1.1.4). Power the setter per 4.6.4.1 while F Control on J7 is grounded. Then attempt to input a "0.1" (E error) via J7 while the continuity loop resistance is greater than 5 ohms. Observe an "F" in the SELECT display for a pass condition.

b. Remove power from the setter and remove the ground on F Control. With the external voltage at the "B" level and the continuity loop resistance greater than 5 ohms, power the setter and attempt to input "0.1" via the test connector. Observe a "B" in the SELECT display for a pass condition.

c. Remove power to the setter and set the external supply voltage to 18.0 volts. With the continuity loop resistance greater than 5 ohms, power the setter and attempt to input a "0.1" via the test connector. Observe a "C" in the SELECT display for a pass condition.

d. Remove power from the setter and return the continuity resistance to no more than 0.1 ohm. Power the setter and attempt to input a "0.1" via the test connector. Observe an "E" in the SELECT display for a pass condition.

4.6.1.8 Fuze memory setting verification and interrogation.

4.6.1.8.1 Fuze counter prejam.

4.6.1.8.1.1 Time prejam. The outputs of Setter Clock lines A, B, and C (J2-S) shall be checked to verify 169 clock pulses are present during prejam, S8 of the setter sequence.

4.6.1.8.1.2 Option prejam. The options 1, 2, and 3, shall be inputted to the setter via J7. The option clock lines shall be checked during S8 of the setter sequence to verify 226 counts for option 1, 258 counts for option 2 and 292 counts for option 3.

4.6.1.8.2 Test/set/verify/sequence. The following tests shall be performed on a fuze (4.6.2) or a simulated fuze (4.5.5) to check the operation of the test/set/verify sequence. A time setting of 13.0 seconds, 70.0 seconds and 125.0 seconds with valid option shall be entered for these tests.

4.6.1.8.2.1 Time setting accuracy. The fuze time memory outputs shall be used to verify that 2-out-of-3 time memories agree within ± 0.075 second of the enter time. The determined time shall be increased by 0.1 second to compensate for fuze battery activation delay.

4.6.1.8.2.2 Option setting accuracy. Each of the valid options shall be entered on the setter, and the fuze memory outputs checked to verify exact agreement with the selected option.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.8.2.3 Time verification accuracy. The time from the FUZE STATUS display shall be compared with the SELECT display. The two displays shall agree to ± 0.1 second.

4.6.1.8.2.4 Option verification accuracy. The option from the FUZE STATUS panel display shall be compared with the SELECT display. The two displays shall agree exactly.

4.6.1.8.2.5 Test/set/verify cycle time. Measure the time required from the detection of a logic high signal on SEQ of the test connector to the loss of fuze power J3-S after S13 has been detected on the test connector.

4.6.1.8.3 Interrogate sequence. The interrogate sequence shall be tested, in the following paragraphs, by entering the keystroke .0 (decimal-zero), and observing the FUZE STATUS display for the appropriate response and a comparison made with the preset parameter to determine pass/fail.

4.6.1.8.3.1 Interrogate time accuracy. Interrogate time accuracy shall be verify by setting the fuze time memory for each channel with a known set time, including 0.1 second for fuze activation delay. The FUZE STATUS display upon interrogate completion shall agree with the set time ±0.100 second.

4.6.1.8.3.2 Interrogate option accuracy. A valid option output shall be provided to the setter when the interrogate sequence has reached S13 as determined on the test connector. The option in the FUZE STATUS display shall compare exactly with the option set.

4.6.1.8.3.2.1 Set 01. Provide set 01 for channels A and B on connector J2-S. Interrogate and verify the displayed option.

4.6.1.8.3.2.2 Set 02. Test as above with a valid set 02A, 02B provided.

4.6.1.8.3.2.3 Set 03. Verify interrogation with a valid set 03A, 03B provided.

4.6.1.8.3.3 Interrogate cycle time. The time required to complete the interrogate sequence shall be tested per 4.6.1.8.2.5 only with interrogate sequence selected.

4.6.1.8.4 Setter time base.

4.6.1.8.4.1 Accuracy. With the setter at 25° ± 5°C, measure the setter time base frequency on the Verify Osc Cont of J7. Compare the reading with 1.024 MHz to determine a pass condition in relationship with the section 3 requirements.

4.6.1.8.4.2 Temperature coefficient. Measure the setter time base on the Verify Osc Cont at -35°C and +55°C. For each case determine the rate of change per °C using the ambient reading of 4.6.1.8.4.1 and compare with section 3.

4.6.1.9 Fuze/setter interface. With the setter interface outputs loaded and the input signals provided per section 4.5.3, the following test shall be performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.9.1 Fuze power. Initiate an interrogate sequence and monitor Fuze Power on J2-S. Measure the time required for the power to the fuze to rise from 3 to 20V dc. Measure the Fuze Power line current 0.5 seconds after the 20 Volt level is reached and verify the compliance to section 3.

4.6.1.9.2 Erase pulse. Power the setter and input a valid time option via the test connector J7. Monitor the Erase/Memorize lines (A on J3-4, B on J3-5 or C on J3-14) during S6 of the setter sequence and measure the pulse width. Also measure the amplitude for the erase pulse at this time. This shall be performed for each Erase/Memorize line. The pulse width and amplitude shall be compared to requirements of section 3. During the pulse duration, Fuze Power on J2-S shall also be measured and verified for compliance to section 3.

4.6.1.9.3 Memorize pulse. Initiate setter power and enter a valid time-option on J7. During S10 of the setter sequence monitor the Erase/Memorize line as in 4.6.1.9.2. Measure the negative going pulse width and amplitude of the memorize pulse verifying that they are in accordance with Section 3.

4.6.1.9.4 Read bias voltage During S13 of a test/set/verify sequence, measure the bias voltage on each Erase/Memorize line at V3-5 pins 4,5, and 14.

4.6.1.9.5 Set Control signal. During a test/set/verify sequence, measure the Setter Control line for each channel, A on J2-S pin 4, B on J2-S pin 14 and C on J2-S pin 8 within 50 ms after the occurrence of S13 on J7. Verify the voltage high level. After MNOS enable during S13 measure the level of the Setter Control line and verify the low level.

4.6.1.9.6 Time set clock signal. Each time set clock shall be monitored, while in a test/set/verify sequence (13.0-1), on J3-S pin 6 Setter Clock A, J3-S pin 15 Setter Clock B and J3-S pin 16 Setter Clock C. In addition the duty cycle for each of the frequency specified shall be measured.

4.6.1.9.6.1 Frequency-time prejam sequence. Measure each setter clock frequency during S8 of the setter sequence.

4.6.1.9.6.2 Frequency-setting sequence. Measure the Setter Clock frequency during S9 of the setter sequence.

4.6.1.9.6.3 Frequency-verification sequence. Measure the Setter Clock frequency during S13 of the setter sequence.

4.6.1.9.6.4 Amplitude. During frequency verification sequence measure the high and low logic levels for each of the Setter Clocks.

4.6.1.9.7 Option set clock signal. The option set clocks, MNOS Enable A J2-S pin 2 and B J2-S pin 3, shall be monitored during a test/set/verify sequence to determine the frequency and amplitudes of the clocks. In addition to the frequency and amplitude, the duty cycle shall be measured. The duty cycle shall be 50% (± 10%).

4.6.1.9.7.1 Frequency. Measure for each clock frequency during S8 of the setter sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.9.7.2 Amplitude. Measure the amplitude of each clock's high and low level during the time specified in 4.6.1.9.6.1.

4.6.1.9.8 Fuze connector resistance. The load resistance on the fuze outputs in the setter shall be measured with a high impedance ohm meter while the setter is not powered.

4.6.1.10 Fuze tests and fault detection. The fuze test and fault detection capabilities of the setter shall be tested by inducing the fuze faults herein. Detection of the faults is done by verifying the appropriate signal on the test connector and FUZE STATUS display. Verification of the FAIL display, sequence inhibit and Fail Disable indications in the test/set/verify sequence shall be performed during the fuze battery test (4.6.1.10.1). Thereafter the Fail Disable shall be sufficient to determine the setter has stopped sequencing and displayed a FAIL in the FUZE STATUS display. The tests shall be performed during a test/set/verify sequence 4.6.1.8.2 or an interrogate sequence 4.6.1.8.3 as required, with a valid time/option entry 4.6.1.5.

4.6.1.10.1 Fuze battery fault. A fuze battery resistance of 500K ohms ± 2% shall be placed across the Battery Mon J3-S pin 7 to fuze ground J3-S pin 3. During the test/set/verify sequence a BA Fail and Fail Disable indication on the test connector shall occur. FAIL on the STATUS display shall also occur and be observed. In addition the Fuze Power shall be monitored for no more than + 1.0V to verify the sequence has been inhibited. A resistance of 1.5M ohms ± 2% shall be placed on the Battery Mon and an interrogate mode initiated. This failure shall produce a BA Fail and Fail Disable on the test connector and a FAIL 4 on the FUZE STATUS display. With resistances of 800K ohms ± 2% and 1.2 Megohms ± 2%, the interrogate mode shall produce a valid time/option display in the FUZE STATUS display.

4.6.1.10.2 Fuze current fault. Apply a 150 ma ± 15 ma load to the Fuze Power lines J3-S pins 2 and 12 during the interrogate sequence. The setter shall display a FAIL 5 indication in its FUZE STATUS display. A logic high signal on Fail Disable and CU of the test connector shall also be detected.

4.6.1.10.3 Fuze clock fault. Each of the three fuze channels shall be tested by the setter in accordance with the clock test matrix during an interrogate sequence. Proper detection of a clock fault by the setter shall occur when a logic high signal on CL of the test connector is detected as indicated in the matrix. A logic high on fail disable of the test connector shall also be detected 1.25 seconds after SEQ on the test connector is a logic high. Determine a FAIL 6 on the setter FUZE STATUS display during one of the tests (other than 10 or 11) in the matrix.

Clock Test Matrix

Test	Clock A Period (ms)	Clock B Period (ms)	Clock C Clock (ms)	CL
1	92	100	100	HIGH
2	100	92	100	HIGH
3	100	100	92	HIGH
4	108	100	100	HIGH
5	100	108	100	HIGH

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

6	100	100	108	HIGH
7	NONE	100	100	HIGH
8	100	NONE	100	HIGH
9	100	100	NONE	HIGH
10	106	106	106	LOW
11	94	94	94	LOW

4.6.1.10.4 Fuze coincidence fault. Fuze coincidence faults shall be detected by the setter when each of the three fuze channels is disabled independent of the remaining channels during sequencing in the interrogate sequence (4.6.1.8.3). Detection is determined by monitoring C1 and Fail Disable on the test connector. Both signals shall be a logic high upon detection. In addition to these detections the test operator shall determine a FAIL 7 indication on the setters FUZE STATUS display. Fuze coincidence windows are 8 counts in the setter. To determine the proper window widths the test connector signals A window, B window and C window shall be tested. These windows occurring when S13 of the test connector is a logic high and have a period of 25 ± 2.5 milliseconds (8 counts at 320 Hz Verify Clk.).

4.6.1.10.5 Fuze accuracy fault. Fuze accuracy fault detection shall be accomplished by providing the setter a verification time which has been either increased by 0.2 seconds or decreases by 0.2 seconds. To accomplish this, all three fuze channels shall have oscillator frequencies that will not cause a one-half cycle round up. This round up occurs when the 2/3 signal in the setter occurs in the last half cycle of the setters 320 Hz Decade Clock (Figure 6). The oscillator frequencies shall be 10270 ± 5 Hz. With these frequencies the set time number of counts shall be increased and decreased by 2 to accomplish the verification errors. The setter shall detect these errors and generate a logic high signal on AC and Fail Disable of the test connector for both increases or decreases in the verification count. The counts used during the verify portion of the test/set/verify sequence shall be used during an interrogate sequence (4.6.1.8.3). Upon completion of the interrogate sequence verify that the FUZE STATUS display of the setter contains the previously set time with the plus or minus two count adjustment.

4.6.1.10.6 Fuze option memory fault. Option memory faults are determined during both a test/set/verify sequence (4.6.1.8.2) and an interrogate sequence (4.6.1.8.3). A logic high on OP Det of the test connector J7 will indicate the detection of an option fault. A logic high on Fail Disable of the test connector will indicate that a FAIL has been generated in the FUZE STATUS display of the setter. The following tests satisfy section 3 requirements upon detection of the logic high state or OP Det and Fail Disable.

4.6.1.10.6.1 Option faults in the test/set/verify sequence.

4.6.1.10.6.1.1 Fault detection in test/set/verify sequence. The following option set signals (4.5.5.5) shall be provided during the test/set/verify sequence. These inputs shall remain present to the setter until the completion of the setter sequencing. A logic high on Opt Detect and Fail Disable on the test connector shall be observed for each test sequence indicating an error has been detected. Each test requires repeating the initiation of a test/set/verify sequence. An X indicates an option set signal shall be generated for that option and channel in performing each test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

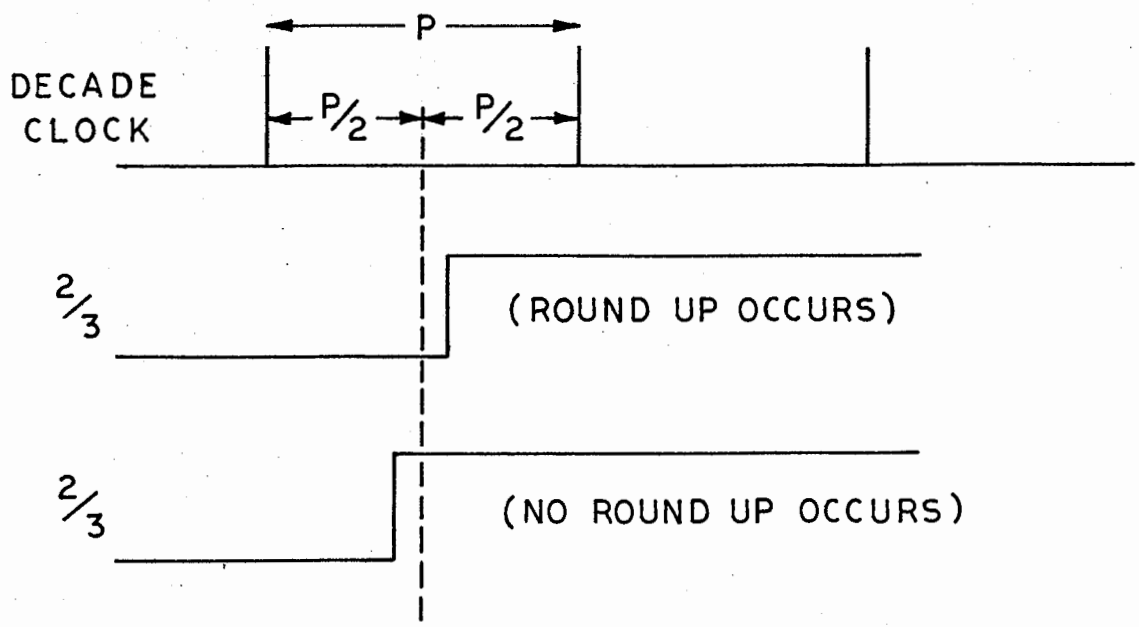


FIGURE 6. ONE HALF CYCLE ROUND UP

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

Test	Select	Channel A			Channel B		
		01	02	03	01	02	03
1	01	X	-	-	-	-	-
2	02	-	X	-	-	-	-
3	03	-	-	X	-	-	-
4	01	-	-	-	X	-	-
5	02	-	-	-	-	X	-
6	03	-	-	-	-	-	X
7	01	X	X	-	X	-	-
8	01	X	-	X	X	-	-
9	01	X	-	-	X	X	-
10	01	X	-	-	X	-	X
11	02	-	X	X	-	X	-
12	02	X	X	-	-	X	-
13	02	-	X	-	X	X	-
14	03	-	X	X	-	-	X
15	03	-	-	X	X	-	X
16	03	-	-	-	-	-	-
17	01	-	X	-	-	X	-
18	01	-	-	X	-	-	X
19	02	X	-	-	X	-	-
20	02	-	-	X	-	-	X
21	03	X	-	-	X	-	-
22	03	-	X	-	-	X	-

4.6.1.10.6.2 Fault detection in interrogate sequence. Repeat test 9 of 4.6.1.10.6.1.1 only this test shall be conducted in the interrogate sequence. In addition to the above acceptance criteria a FAIL 8 in the setter FUZE STATUS display shall be observed.

4.6.1.11 Induced fault safety requirements.

4.6.1.11.1 Fuze battery detection circuit fault. BA1 Disable and BA2 Disable are used to selectively disable each of the two redundant fuze battery test circuits. Applying a ground to BA1 Disable and repeat 4.6.1.10.1 the fuze battery fault shall be detected by the setter. Repeat the above for BA2 Disable grounded and BA1 Disable open circuit. The fuze battery fault shall be detected.

4.6.1.11.2 Time 2/3 decode logic fault. 2/3 CONT A and 2/3 CONT B lines on the test connector provide access to the setter's 2/3 detection circuitry. By selectively grounding each of these lines each of the redundant 2/3 test circuits can be disabled. Apply a ground to 2/3 CONT A. Power the setter (4.6.4.1) and initiate a valid time and option entry (4.6.1.5). A Fail Disable shall be detected on the test connector. Repeat the above with 2/3 CONT B grounded and 2/3 CONT A open circuit. Again a Fail Disable on the test connector shall be present.

4.6.1.11.3 Time accuracy error detection fault. With the fuze oscillator frequencies set to 10270 ± 5 Hz, power the setter (4.6.4.1) with the Fail Disable line of the test connector grounded. Enter a valid time and option (4.6.1.5). Upon determining the set time from the setter decrement all channels (4.5.5.10) by 15 counts. Upon completion of the setter sequencing, determine the time displayed in the setters FUZE STATUS display. The time observed shall be 1.5 ± 0.1 seconds less than the selected time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.1.11.4 Setter time base fault. Setter time base faults shall be induced by applying a low (50 ohms or less) impedance source to the Verify Osc Cont of the test connector. This source shall be 5 ± 0.2V for a logic high, less than 1.0V for a logic low and have a duty cycle of 50% (± 10%). To induce the faults a frequency of 1.11 MHz and 0.942 MHz shall be provided to the setter during an interrogate sequence (4.6.1.8.3). A fuze clock test failure shall be detected for each of two frequencies when a 100 ms ± 2% period is provided for each of the fuze clock inputs. Detection of this failure shall be accomplished by monitoring the CL line of the test connector for a logic high signal.

4.6.1.11.5 Option error detection fault. The option error detection circuit FAIL indication shall be disabled by grounding the Fail Disable signal on the test connector. The setter shall be powered (4.6.4.1) and a valid time and option entry made. During the setters test/set/verify sequence the option set inputs (4.5.5.5) listed below shall be provided to the setter. Determine that the resultant FUZE STATUS display of the setter agrees with the table below for a pass condition.

Option Sets

SET	Channel A			Channel B			FUZE STATUS Option Digit
	01	02	03	01	02	03	
01	X	X	-	X	X	-	2
01	X	-	X	X	-	X	3
02	-	X	X	-	X	X	3
02	X	X	X	X	X	X	3

X indicates the presence of an option set input.

4.6.2 Fuze setting test.

4.6.2.1 Fuze setting.

4.6.2.1.1 Production acceptance fuze setting. Two M735 fuze nose sections shall be set to the settings listed below. Verify the setter acceptance per 3.3.1.8.2.3, 3.3.1.8.2.4, 3.3.1.8.3.1 and 3.3.1.8.3.2. After the nose sections have been set, the time and option settings shall be verified by an external counter with the nose section powered and operated without any other external stimulus. The externally verified time and option setting shall be compared to the requirements of 3.3.1.8.2.1 and 3.3.1.8.2.2 for acceptance. The externally verified time shall be the 2-out-of-3 decoded time output from a 120 count decode counter provided externally. In addition the external time shall be adjusted by 0.100 seconds to account for fuze battery rise time.

Time	Option
13.0	-1
70.0	-2
125.0	-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.2.1.2 FAAS fuze setting. Perform tests described in 4.6.2.1.1 with five instead of two M735 nose sections.

4.6.2.2 Fuze memory retention.

4.6.2.2.1 FAAS test. After completion of the 125.0 - 3 setting of 4.6.2.1.1, the fuze nose section shall be left unpowered and at ± 3°C of its set temperature for a minimum of 24 hours. Upon completion of the 24 hour period, verify the setter displays the nose section setting of 125.0 ± 0.1 seconds with a 3 option in the interrogate sequence (4.6.1.8.3). In addition, verify the fuze set time and option on external playback equipment as in 4.6.2.1.1

4.6.2.2.2 Production acceptance test. Perform the test specified in 4.6.2.2 except change the interval between setting and interrogation from 24 hours to 1 hour.

4.6.3 Human factor test.

4.6.3.1 Keyboard contact force. Using a push gauge, measure each key contact force for each of the following fuze settings. Key force shall not exceed the requirements of section 3.

Time	Option
45.6	1
78.9	2
123.0	3

4.6.3.2 Keyboard debouncing. Using a linear solenoid apply a 14 oz. pressure over a 0.1 inch-travel in 14 ms to each of the key 10 numeric entries. The pressure shall be maintained for 50 ms and returned to 0.0 oz of pressure in 14 ms. This shall be repeated one more time after a 50 ms pause. No false entries on the SELECT display (three or more) shall be detected.

4.6.3.3 LED brightness. Using a Tektronix J16 photometer with a J6503 probe or equivalent, measure each LED digit of the SELECT and FUZE STATUS displays. Each display shall be compared to section 3 for acceptance.

4.6.3.4 Blackout keyboard visibility. Measure each of the 6 keyboard LED lamps using the instrument specified in 4.6.3.3. The LED's shall have an 0.6 foot lamberts output.

4.6.3.5 Display Functions. The following functions shall be checked during a test/set/verify sequence 4.6.1.8.2 with a valid time option entry 4.6.1.5 unless otherwise specified. The verification of each test shall be determined when the observation of the specified response from the setter is present.

4.6.3.5.1 Setter busy signal. A flashing digit in the FUZE STATUS display shall be observed in the interval between the last keyboard entry from the test connector and memory playback sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.3.5.2 Status display retention. When the verification sequence is completed, 4.6.1.8.2.3 and 4.6.1.8.2.4, the FUZE STATUS display shall be observed to verify the display remains unaltered until setter power is turned off or a new entry has been completed.

4.6.3.5.3 Time option dash. During the time/option entry, observe that the dash separting time and option on the SELECT display does not display until the last time entry is made. While in the verification sequence, the dash on the FUZE STATUS display shall not be on until the valid time has been displayed.

4.6.3.5.4 Decimal point sequence. When the setter is first powered up with a CE entry, observe the SELECT display to verify that a 0. is displayed. A numeric keyboard entry shall remove the decimal point and the selected keyboard input be displayed. The decimal point shall be displayed again when a decimal point is keyed into the setter.

4.6.3.6 Fuze insertion-removal. An acceptable fuze nose section (MIL-F-48703) shall be installed on a setter using a push-type gauge to determine the force required to obtain an adequate setting. A time/option setting 13.0 option 1 shall be used to determine this condition. Similarly, with the setter on after the completion of a test/set/verify sequence, a pull-type gauge shall be used to determine the force required to obtain a C in the setters SELECT display. These readings shall be used to determine the units acceptance.

4.6.3.7 Setter weight. Using a calibrated scale, measure a completed setter and compare to the requirements at section 3.

4.6.4 Power supply test.

4.6.4.1 Setter power control. The CE entry key shall be checked to verify that one keystroke will power the setter and two CE keystrokes will remove power from the setter.

4.6.4.2 Automatic turn-off. Turn-off time shall be measured by initiating a power-on and monitoring F Control of the test connector to determine when the setter power is turned off. The elapse time shall be measured and compared to section 3 for a pass/fail acceptance.

4.6.4.3 Power cable requirements. The following test shall be accomplished on a completed power cable assembly 11711399.

4.6.4.3.1 Resistance. Measure the voltage drop across the cable which is conducting 1 amp of current. The voltage drop shall not exceed 0.35V on the center contact or the outside contact.

4.6.4.3.2 Current. Apply a 3.0A current to J1 with the trouble light connector shorted. The voltage drop across J1-A, J1-B shall be no greater than 2.1V. After measuring this voltage drop, increase the current to 40.0A for 50 ms. Again measure the voltage drop across the two connector pins. The voltage drop shall be no more than 0.1 volt from the previous reading. The second reading shall be made 5 minutes after the high current pulse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.4.3.3 Isolation. Measure the isolation between J1-A and J1-B using a 500V dc source. Readings shall be compared to section three requirements for a pass/fail indication.

4.6.4.4 Power indicator. The test shall verify that the keyboard LED lamps are activated when the external power is applied to J1-S in either power-on or power-off modes.

4.6.4.5 EMR rejection filter. The shield case assembly (11735239) shall be tested with the chassis assembly (11735215) removed. The input to the filters (outside of case) shall be powered with 33.0 ± 0.3V dc while the outputs of the filter are loaded with 16.5 ohm ± 1.0% resistor, 70 watts minimum. Superimpose a 30 KHz signal on the input lines and measure the resultant signal at the load. Repeat with a 150 KHz signal. Compare the results of these tests with section 3 requirements for a pass/fail indication.

4.6.5 Environmental tests. Unless otherwise specified, the complete setter assembly (PN 11727881) shall be subjected to the environments specified below.

4.6.5.1 Air temperature extremes. Environment No. 1 of 11710169.

4.6.5.2 Temperature-humidity. Environments No. 2 and 3 of 11710169.

4.6.5.3 Thermal shock. Environment No. 4 of 11710169.

4.6.5.4 Pressure. Environment No. 5 of 11710169.

4.6.5.5 Salt fog.

4.6.5.5.1 Salt fog. Environment No. 6A of 11710169.

4.6.5.5.2 Salt fog (safety). Environment No. 6B of 11710169.

4.6.5.6 Sand and dust.

4.6.5.6.1 Sand and dust. Environment No. 7A of 11710169.

4.6.5.6.2 Sand and dust (safety). Environment No. 7B of 11710169.

4.6.5.7 Rain.

4.6.5.7.1 Rain. Environment No. 8A of 11710169.

4.6.5.7.2 Rain (safety). Environment No. 8B of 11710169.

4.6.5.8 Fungus. Environment No. 9 of 11710169.

4.6.5.9 Seal. Environment No. 10 of 11710169.

4.6.5.10 Surface and air transportation vibration.

4.6.5.10.1 FAAS vibration. Environment No. 11A of 11710169.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

4.6.5.10.2 Production acceptance vibration. Environment No. 11B of 11710169.

4.6.5.11 Impact.

4.6.5.11.1 Side drop. Environment No. 12A of 11710169.

4.6.5.11.2 Cornerwise drop. Environment No. 12B of 11710169.

4.6.5.11.3 Edge drop. Environment No. 12C of 11710169.

4.6.5.11.4 Production acceptance drop. Environment No. 12D of 11710169.

4.6.5.12 Ice fog. Environment No. 13 of 11710169.

4.6.5.13 EMR. Environment No. 14 of 11710169.

4.6.5.14 Lightning EMP. Environment No. 15 of 11710169.

4.6.5.15 Static electricity. Environment No. 16 of 11710169.

4.6.5.16 Transient radiation. Environment No. 17 of 11710169.

4.6.5.17 EMP, nuclear. Environment No. 18 of 11710169.

4.6.5.18 Life. Environment No. 19 of 11710169.

5. PACKAGING.

5.1 Cleaning, preservation and packaging. The cleaning, preservation and packaging of items for protection against corrosion, physical damage, and other forms of deterioration during handling, shipment, and storage, shall be in accordance with MIL-P-116.

5.2 Packing. The assemblies shall be prepared for delivery in accordance with good commerical practices to protect the parts against any damage during direct shipment form the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers, shall be marked in accordance with requirements of MIL-STD-129.

5.4 Records. The copy of the data (4.4.3) shall be shipped as directed by the procuring activity. The master of the original of the data shall be retained by the contractor for the duration of the contract.

6. NOTES

6.1 Items to be included in the procurement documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

6.1.1 Documents. Procurement documents should specify the title, data, and number of the specification.

6.1.2 Samples. The number of units should be increased to include the samples as required in 3.1.1 and 3.1.3, since they are not included in the number of units required for delivery.

6.1.3 Government testing of comparison samples. These samples are subject to retesting by the Government to insure that first article approval requirements continue to be met. Failure of these samples will not affect the acceptance of the lot, but may indicate future mandatory action.

6.1.4 Lead time. The testing of the first article approval sample will require a maximum of 60 working days. If there are failures, additional time will be required.

6.1.5 Data requirements. The following data are required by this specification to be furnished to the Government and should be entered on DD Form 1423, if made a part of the contract.

First Article Test results (4.3)
Comparison Test results (4.4.5.3)
Log books containing production acceptance test results (4.4.3.1)
MIL-Q-50829 paragraph 3.6.1 - Contractor-furnished designs

6.1.6 First Article Approval. The contract will specifically state those tests conducted by the Government Technical Agency in accordance with Note 1 of paragraph 4.4.5.2 that are to be mandatory for first article approval.

6.2 Advisory documents. The following documents are not part of this specification, but may provide useful information to the contractor. Copies are available for reference at the Harry Diamond Laboratories.

6.2.1 Test equipment manuals. (not available)

6.2.2 Quality assurance system documents (obtained from previous contracts).

6.2.3 Stockpile to target sequence (STS).

6.3 Engineering information. Additional data beyond that required for production acceptance or data requiring special requirements, when needed for engineering purposes, will be specifically defined by the contract.

6.4 Storage life. The design objective regarding setter storage life is that the setter be capable of full performance after 5 years storage with no maintenance. In addition the setter design and components should be capable of surviving 24 years storage with test and repair intervals not less than 5 years. The storage environment is not expected to be more severe than the environments specified in 3.3.5.

6.5 Safety design. The detrimental safety effects resulting from potential circuit faults in delivered setters are substantially minimized by utilizing an intelligent FUZE STATUS output display (as opposed to a pass/fail type)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.

which allows the operator to discern incorrect setting results even if internal fault detection circuits or critical setting functions fail. In addition, redundant fault detection circuits are provided in those areas where the setter display is insensitive to a specific setter circuit fault. The specifications in paragraph 3.3.1.11, "Induced fault safety requirements", are intended to insure that the setter's FUZE STATUS display reacts to potential critical setter faults in a safe manner, and that those fault detection circuits which are redundant are fully operable.

Preparing Activity
Army-AR
Project No: 1390-A375

Source: https://assist.dla.mil/ -- Downloaded: 2026-10-22T19:37Z
Check the source to verify that this is the current version before use.